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Topic

BPDZ4APA : A Centralized InterBank Platform for Algerian Payement Area

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Rèsumè

Ce mémoire propose la conception d'une plate-forme de paiement interbancaire nationale baptisée BPDZ, inspirée des systèmes européens SEPA et TARGET2. Cette solution vise à moderniser les échanges financiers en Algérie, à réduire l'utilisation du cash et à dynamiser le marché économique national. Bien que l'Algérie dispose déjà de systèmes RTGS, ceux-ci restent insuffisants pour répondre aux exigences de l'économie moderne et de l'intégration régionale.

L'objectif est de faire de la banque un acteur central dans le développement économique à travers un système de paiement structuré, fiable et interopérable. Le recours à la norme ISO 20022 permettra à la plate-forme d'être conforme aux standards internationaux, rendant ainsi BPDZ potentiellement éligible à une reconnaissance régionale et mondiale.

Le projet prévoit aussi l'extension de cette solution à un espace économique maghrébin, puis africain, à travers la création de l'espace APA (Espace de Paiement Algèrien), analogue au SEPA mais adapté aux spécificités régionales. Ce mémoire met en lumière les enjeux technologiques, économiques et stratégiques d'une telle transition, tout en offrant une vision prospective vers l'intégration financière africaine.

Abstract

This report proposes the design of a national interbank payment platform named BPDZ, inspired by the European SEPA and TARGET2 systems. The initiative aims to modernize financial transactions in Algeria, reduce reliance on cash, and energize the domestic economic market. Although Algeria has existing RTGS infrastructures, they remain insufficient for meeting the demands of a modern, integrated economy.

The project seeks to position banks at the core of a structured, secure, and interoperable financial system. By adopting the ISO 20022 standard, the platform will align with international norms, making BPDZ a potential candidate for regional and global compatibility.

Furthermore, the solution envisions the creation of a broader Algerian Payment Area (APA), a regional space inspired by SEPA but tailored for the Maghreb and African contexts. This report emphasizes the technological, economic, and geopolitical implications of the platform, offering a strategic vision for regional financial integration.

ملخص

تقترح هذه المذكرة تصميم منصة وطنية للدفع بين البنوك تُدعى BPDZ، مستوحاة من نظامي SEPA وTARGET2 الأوروبيين. تهدف هذه المبادرة إلى تحديث المعاملات المالية في الجزائر، وتقليل الاعتماد على النقد، وتنشيط السوق الاقتصادية المحلية. وعلى الرغم من وجود بنية تحتية للدفع الإجمالي اللحظي (RTGS) في الجزائر، إلا أنها لا تزال غير كافية لتلبية متطلبات اقتصاد عصري ومتكامل.

يسعى المشروع إلى وضع البنوك في قلب نظام مالي منظم وآمن وقابل للتشغيل البيني. ومن خلال اعتماد معيار ISO 20022، ستتوافق المنصة مع المعايير الدولية، مما يجعل من BPDZ مرشحًا محتملاً للتكامل الإقليمي والدولي.

علاوة على ذلك، تتطلع هذه الحلول إلى إنشاء فضاء أوسع يُعرف باسم منطقة الدفع الجزائرية (APA)، وهو فضاء إقليمي مستوحى من SEPA، لكنه مصمم خصيصًا ليتماشى مع السياقين المغاربي والإفريقي. تسلط هذه المذكرة الضوء على الأبعاد التكنولوجية والاقتصادية والجيوسياسية للمنصة، وتقدم رؤية استراتيجية للتكامل المالي الإقليمي.

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1 General Introduction

In any modern economy, the role of banks extends far beyond the simple management of deposits and loans. Banks are a fundamental pillar of economic development, ensuring the smooth flow of financial transactions and maintaining the trust of economic stakeholders.

In Algeria, as in many other countries, the banking sector should play a central role in structuring and developing economic networks. This centrality is particularly crucial when it comes to interbank transactions, fund transfers, and especially customer payments whether involving businesses or individuals.

However, in a context where the economy is still heavily dependent on cash, the limitations of this mode of transaction are becoming increasingly evident:

- Lack of transparency in financial flows.
- Security risks.
- High management costs.
- Difficulty for the state and investors to obtain a reliable picture of the market.

In response to these challenges, many countries in Europe, Asia (notably India), and Africa have begun deeply transforming their banking infrastructure by establishing **centralized interbank platforms**, often under the authority of their central banks.

These platforms aim primarily to provide:

- **Security**
- **Interoperability**
- **Irrevocability**
- And **cost optimization** for interbank transactions

By offering these guarantees, such systems not only strengthen trust among financial institutions but also encourage clients both businesses and individuals to rely on banking services, thereby reducing dependence on cash. As a result, **modern payment systems** become a strategic tool to:

- Formalize the economy.
- Improve the transparency of financial flows.
- Foster healthier and more sustainable economic growth.

This report therefore aims to explore, from both a **comparative and technological perspective**, how the implementation of a national interbank platform could address these challenges in Algeria, and what **technical, functional, and institutional foundations** would be necessary to ensure its success.

2 Problem Statement

2.1 Challenges of Interbank Transactions in Algeria

Despite showing solid economic performance in 2023 with a 4.1% GDP growth, Algeria's economy remains heavily dependent on hydrocarbons. While the financial system is gradually evolving, including efforts in digitalization and the introduction of Islamic finance, several **structural weaknesses persist**, particularly in the **interbank transaction system**.

Although the use of electronic payments is growing (cards, mobile apps like BaridiMob), **cash usage remains dominant**, especially among individuals. Access to banking services in remote areas is still limited, and public trust in banks is not yet fully established, which slows the adoption of digital payments.

Algeria's current interbank infrastructure composed mainly of **a centralized clearing system managed by the central bank** and **SATIM** for card operations lacks the **real-time capabilities** and **client-to-client payment fluidity** offered by modern platforms found in Europe (SEPA) or India (UPI).

2.2 Limitations of the Current System

- There is **no unified national interbank platform** capable of processing real-time transactions between banks.
- Payment infrastructure is **fragmented**, with each bank relying on its own internal systems.
- Current platforms are **insufficient for fast, secure, and accessible client-to-client interbank payments**.

2.3 Need for a Unified and Standardized Solution.

There is a growing need in Algeria for a centralized interbank platform that ensures:

1. Real-time and secure settlements.
2. Interoperability between all banks.
3. Compliance with international standards, such as ISO 20022.
4. Improved liquidity management via central bank money.

2.4 Main Problem

How can a centralized, standardized, and secure interbank platform aligned with ISO 20022 help modernize Algeria's financial infrastructure and overcome the limitations of the current fragmented banking system?

3 Report Structure

Chapter 1: Context and Literature Review

This chapter presents the current challenges facing the Algerian banking sector and introduces the motivation for developing a modern, secure, and interoperable payment solution tailored to Algeria's financial context. It also provides an overview of interbank payment systems such as SEPA, SWIFT, and international standards like ISO 20022.

Chapter 2: Modeling and Process Workflows

This chapter details the system design using UML modeling. It includes the following diagrams:

- **UML Overview** : Introduction to the modeling language used.
- **Use Case Diagram** : Identification of system actors and their interactions.
- **Class Diagram** : Structural representation of the system components.
- **Sequence Diagram** : Dynamic flow of messages and use case scenarios.

Chapter 3: Implementation and Evaluation

This chapter discusses the technical realization of the system and evaluates its performance:

- **Implementation and Development** – Technologies, tools, and steps involved in building the solution.
- **Security Aspects** : Security measures integrated into the system.
- **Testing, Results** : Testing methodology, results obtained.
- **Conclusion,and Perspectives** : Summary of findings and final remarks and Perspectives and possible future improvements.

CHAPTER 1 : Context and Literature Review

1 Introduction

This chapter explores Algeria's current interbank systems, highlights international models, and emphasizes the need for a modernized solution suited to Algeria's needs . And the return on investment of the proposed system .

2 Context and Literature Review

2.1 Overview of International Interbank Systems

2.1.1 SWIFT Network

SWIFT, created in 1973, is a secure global messaging system for international bank transfers. Algerian banks use it for cross-border transactions. It handles over **45 million messages daily**, but its vast scale also makes it vulnerable to **money laundering risks**, estimated at **$2 trillion USD annually**.

It is important to note that:

- **SWIFT is not a payment processor** : it only transmits messages (no money is moved, no accounts are held).
- **Using SWIFT is not free** : banks are charged based on their transaction volume.

2.1.2 Interbank Exchange Plateforms:

- **SEPA/TARGET2 (Europe)**: TARGET2 is a real-time gross settlement (RTGS) system operated by the European Central Bank. It ensures fast, irrevocable, and secure interbank payments in euros. SEPA complements this by harmonizing payments across the eurozone, allowing consumers and businesses to make cross-border transfers under the same conditions as domestic ones. SEPA also fully adopts the ISO 20022 standard.
- **Fedwire (USA)**: An RTGS system run by the U.S. Federal Reserve, used for high-value, urgent domestic transactions. It offers secure, final settlement between financial institutions holding accounts with the Federal Reserve.
- **UPI (India)**: A modern, API-driven system that enables instant payments between bank accounts 24/7. Managed by the National Payments Corporation of India, UPI has revolutionized retail payments and reached over **10 billion monthly transactions** by 2024, promoting financial inclusion and reducing reliance on cash.
- **GIM-UEMOA (West Africa)**: A regional platform connecting banks from 8 countries under a shared payment system. It enables interoperable services like card payments, ATM access, and mobile transactions, supporting regional trade and banking integration.

2.2 The Algerian Interbank Landscape:

Algeria has foundational infrastructure for interbank payments, managed by the **Bank of Algeria**, including:

- **RTGS** for high-value, real-time settlements;
- **ACH (ATCI)** for mass transactions like checks and salary transfers;
- **SATIM** for card-based payments (CIB cards, ATMs, POS systems).

Some banks, such as the **Banque Nationale d'Algérie (BNA)**, have developed their own platforms that connect to RTGS and support API-based services, but these remain siloed and not interoperable with other banks.

2.2.1 Current Limitations and Future Outlook:

Despite existing efforts, the current approach remains fragmented, with each bank operating its own technical solutions. There is still no **unified and open interbank platform** that enables seamless interoperability between all commercial banks in Algeria.

The main limitations of the Algerian interbank system include:

- The absence of a **centralized exchange hub** or single access point.
- **Technological dependence** on proprietary or internal systems specific to each bank.
- A **lack of standardized exchange formats** across the sector.

To address these challenges, several initiatives are being considered:

- **Strengthening the adoption of ISO 20022** standards throughout the banking sector.
- Developing an **open national interbank platform**, supervised by the **Bank of Algeria**.
- Increasing **collaboration between banks** to establish shared infrastructure in a spirit of unification and efficiency.

These developments are crucial to support the **digitalization of the Algerian economy**, reduce transaction costs, improve transparency, and prepare for **potential regional integration** (with the Maghreb or the African Union).

2.2.2 Weaknesses and Opportunities:

Algeria's interbank system still faces several key limitations:

1. No **real-time interbank transfers** or **open banking APIs**.
2. No **real-time client payment platform** like RTGS or UPI.
3. Limited **interoperability between banks and fintechs**.
4. Lack of **24/7 instant payment availability** for individuals and businesses.
5. No strong integration with **fiscal or commercial digitalization** to reduce cash usage.
6. No **clear strategic vision** yet published by the central bank for a unified model.

In summary, while Algeria has a solid foundational system led by the central bank and key players like BNA, the shift toward a **more integrated, interoperable, and digital platform** remains a major strategic challenge for the coming years.

2.3 What Algeria Can Learn from Other Systems:

To develop a sovereign and modern interbank payment platform inspired by systems like **TARGET2** or **SEPA**, Algeria must adopt a set of proven **best practices**, robust **technologies**, and a clear **strategic vision**. The following summarizes the key principles and lessons:

2.3.1 Best Practices and Key Technologies:

1. Transaction Security Any interbank platform must prioritize strong multi-layered security:

- **Data encryption** (e.g., TLS 1.3, AES-256).
- **Strong authentication** (digital certificates, HSM tokens, OTP).
- **Traceability** (immutable logs, timestamped actions).
- **Fraud detection** (rule-based systems or AI).

2. System Resilience and Availability

- **Redundant servers and databases** with automatic failover.
- **Disaster recovery plans (DRP)**.
- **Real-time system monitoring and alerts**.

3. **Interoperability and Standardization**
   - Use of **ISO 20022** (e.g., pacs.008) for message formats.
   - **Secure RESTful APIs** for communication between banks.
   - Standardized **XML/JSON schemas** for financial data.
4. **Governance and Compliance**
   - Supervision by the **Bank of Algeria**.
   - Complete **auditability and documentation**.
   - Strict adherence to **data protection** principles.
5. **Institutional User Experience**
   - **Clear user portals** for operators and supervisors.
   - **Real-time dashboards** for tracking transactions and balances.
   - **Role-based access control** for different user profiles.

2.3.2 Key Takeaways for Algeria:

- A **national interbank platform**, backed by the **Bank of Algeria**, could follow the **TARGET2 model** for real-time payments, while incorporating **SEPA-like standards** for interoperability.
- Collaboration between **banks, telecom operators**, and the **central bank** could support a **mobile-first user experience** that competes with cash usage.
- This platform could also serve as a **foundation for regional cooperation**, enabling cross-border payments with neighboring countries like Tunisia, Morocco, and Mauritania.

2.4 Conclusion of the Context and Literature Review:

While Algeria has some technical infrastructure in place (e.g., SATIM, ACH, bank connectivity), the **main obstacles are institutional and strategic**. The lack of a unified national platform slows down the digitalization of payments and limits financial inclusion.

Building such a platform is **not just a technical challenge** but a **strategic opportunity** to:

- Modernize the economy.
- Increase transparency.
- Strengthen banking systems, and
- Reduce reliance on cash.

With the right vision, Algeria can build a **resilient, inclusive, and sovereign interbank platform** that repositions its financial system at the core of national economic activity.

3 Presentation of the Proposed Platform:

3.1 Introduction to the APA Platform Subject of Our Report:

The **APA platform (Algerian Payment Area)** proposed in this report aims to serve as a foundation for the modernization of Algeria's banking infrastructure. Banks and financial institutions play a key role in regulating and energizing the market by facilitating the flow of monetary and financial transactions. However, they must be equipped with the right tools and infrastructure to participate effectively and take a leading role in this transformation.

3.2 Banks Need Performance and Guarantees:

The need for high performance has become essential due to the development of financial markets and the increasing specialization of banks (retail banking, investment, asset management, etc.). The growing volume of transactions requires **intelligible** and **interoperable** digital connections between these institutions.

Note: In this context, *intelligible* and *interoperable* mean that a clearly defined and readable protocol is shared in advance between the sender and receiver, even if they operate different digital systems. This also includes interactions between banks in different countries.

3.3 Why a Centralized Payment Platform is Critically Useful:

One of the key roles of a **central bank** is to ensure the smooth operation of the interbank market, particularly by providing liquidity to banks for their financial transactions. In return for services (account management, payment tools, financial advice, etc.), banks earn commissions from their clients.

A **centralized interbank platform** managed by the **Bank of Algeria** would offer the following benefits:

- **Fast and secure settlements**: Payments are processed in real-time, reducing liquidity and counterparty risk with a high level of security.
- **Facilitated cross-border and interbank transactions**: Especially within the APA zone (Algerian Payment Area).
- **Improved liquidity**: Through immediate settlement using central bank money.
- **Reduced credit and settlement risks**: By preventing the accumulation of receivables between banks. Centralized payments also reduce costs and improve banks' cash management.
- **Regulatory compliance**: Central bank management ensures adherence to standards and financial stability.
- **Strengthened financial stability and monetary policy**: A centralized platform allows better liquidity control and more effective transmission of monetary policy. By centralizing financial flows, the Bank of Algeria could better monitor and adjust monetary conditions, reducing systemic risks.
- **Optimized reserve requirements and money market efficiency**: With Algeria's current reserve requirement set at 3%, a centralized system would improve reserve management and support refinancing or investment operations. This would boost interbank liquidity and lower financing costs.
- **Operational efficiency for banks**: Standardizing payment processes would cut operational costs and processing time, allowing banks to focus more on innovation and competitive services.
- **Increased foreign investment**: A transparent and efficient payment system would enhance investor confidence, facilitating smoother transactions and risk management, thus attracting more foreign direct investment (FDI).
- **Regional integration and economic development**: The platform could form the basis for integrating payment systems across the **Maghreb and Sub-Saharan Africa**, promoting intra-African trade and positioning Algeria as a regional financial hub.

3.4 Return on Investment (ROI) Short, Medium, and Long Term:

3.4.1 Basic Financial Assumptions (in Algerian Dinar DZD):

Item	Estimated Value
Initial investment (infra, licenses, HR...)	800 to 2,000 million DZD
Expected annual revenue at maturity	600 to 1,200 million DZD
Break-even period	3 to 5 years
Exchange rate (indicative)	1 USD 135 DZD

The ROI estimates are based on:

- Costs observed in similar African projects.
- National economic volumes from public reports.
- International benchmarks on centralized payment systems.
- A realistic 10-year financial model.

Period	Main Expenses/year	Potential Revenue/year	Estimated ROI
Short-term (1 to 3 yrs)	800–1,500 M DZD	50–100 M DZD	-20% to -50%
Medium-term (3 to 6 yrs)	150–300 M DZD	250–600 M DZD	+15% to +40%
Long-term (7 to 15 yrs)	100–200 M DZD	800–1,200 M DZD	+70% to +200%

3.4.2 Recommendations to Maximize ROI:

- Initial **launch subsidies** by the **Bank of Algeria** to absorb startup costs
- Rapid onboarding of large public enterprises (e.g., **SONELGAZ**, **ALGÉRIE TÉLÉCOM**, **CNAS**)
- Use the platform for **public aid, pensions, and salary payments**
- Develop **integrated financial services** (e.g., microcredit, deferred payments)
- Partner with **postal services or fintechs** to promote rural financial inclusion

3.5 Integrating ISO 20022 Standards into BPDZ:

ISO 20022 is a global messaging standard that enhances. **interoperability, automation, and data richness** in financial transactions. It uses **XML format** to structure messages consistently across payment, securities, and treasury systems.

The BPDZ platform adopts ISO 20022 to ensure **standardization, compliance, and real-time financial transparency**. It supports the following key message types:

3.5.1 Supported ISO 20022 Messages in BPDZ:

- **PACS.008**: Credit Transfer : used for transferring funds between bank accounts.
- **PACS.009**: Financial Institution Transfer : for high-value interbank transactions (replaces SWIFT MT202).
- **PACS.002**: Payment Status Report : acknowledges receipt, acceptance, or rejection of a payment.
- **PACS.003**: Direct Debit Request : initiates a debit based on a mandate (e.g., utility bills, loans).
- **PACS.004**: Payment Return : reverses a transaction due to failure or rejection.
- **CAMT.053**: Bank Statement : provides detailed, real-time reports of account activity and balances.

3.5.2 Benefits for BPDZ:

- **Automation & Error Reduction**: Rich, structured data reduces manual intervention.
- **Real-Time Processing**: Supports instant payments, settlement tracking, and transparency.
- **Global Compliance**: Aligns BPDZ with international systems like SEPA, SWIFT, and TARGET2.
- **Efficient Reconciliation**: CAMT.053 enables accurate and up-to-date financial reporting.

By integrating ISO 20022, BPDZ ensures **future-proof compliance, operational efficiency, and global interoperability**, positioning itself as a modern, secure, and transparent banking platform.

4 Conclusion

The creation of a centralized interbank platform under the supervision of the central Bank of Algeria would offer multiple advantages, ranging from increased financial stability to the attraction of foreign investment. By modernizing the national financial infrastructure, Algeria would be better positioned to meet current and future economic challenges.

5 Glossary of Technical and Financial Terms:

Table 1.1: **Key Financial and Interbank Concepts**

Term	Description	Usage in Algeria	Global Usage
IBAN	International Bank Account Number used to uniquely identify accounts across borders.	Used for cross-border transactions via SWIFT.	Standard for cross-border payments worldwide.
RTGS	Real-Time Gross Settlement system for immediate settlement of high-value payments.	Used by the Bank of Algeria for interbank payments.	Core to systems like TARGET2 (EU), Fedwire (US).
DNS	Deferred Net Settlement batch processing system settling net positions later.	Used in domestic clearing (e.g., ATCI).	Common for mass payments in ACH systems.
Loans	Money lent to individuals or institutions to be repaid over time with interest.	For consumer, SME, and mortgage financing.	Universal tool for economic growth and investment.
Mandate	A pre-approved authorization for direct debit operations.	Common in utilities and recurring payments.	Fundamental to SEPA Direct Debit operations.
BIC	Bank Identifier Code (SWIFT Code) identifies banks in international payments.	Required for SWIFT wire transfers.	Global standard in cross-border transactions.

Table 1.2: **ISO 20022 Message Types Integrated in BPDZ**

Message Type	Role	Description
PACS.002	Payment Status Notification	Acknowledges a payment's status (accepted, rejected, pending).
PACS.003	Direct Debit Request	Initiates a debit from a debtor's bank based on an agreed mandate.
PACS.004	Payment Return	Reverses a previously processed payment due to errors or rejections.
PACS.008	Credit Transfer	Standard interbank payment between two accounts.
PACS.009	High-Value Interbank Transfer	For large settlements between financial institutions.
CAMT.053	Bank Statement	Provides a full transaction history and balances in real-time.

Table 1.3: **Project-Specific Terms**

Term	Description
APA (Algerian Payment Area)	The national interbank platform proposed in this report, designed to modernize Algeria's banking system. Inspired by TARGET2/SEPA, APA supports ISO 20022 messaging, real-time payments, regulatory compliance, and regional interoperability.
BPDZ (Banking payment DZ)	A digital simulation of an interbank messaging platform used in this report. It uses a unified database model, Spring Boot backend, and ISO 20022 messages for testing and development of the APA system.

CHAPTER 2 : Modeling and Process Workflows

1 Introduction :

This chapter outlines the design of our BPDZ (Banking Platform DZ) application, starting with its objectives. We will then describe the technical architecture and the various functionalities of the platform using UML diagrams. Our analysis will include use case scenarios to identify the relevant classes and the sequence of flows for each implementation. The goal is to provide a clear and structured overview of how the platform operates, ensuring a seamless integration of financial services in Algeria, while adhering to global standards.

2 Definition of UML :

The Unified Modeling Language (UML) is a graphical language used to visualize, specify, and design systems, particularly in object-oriented development. It was created by merging earlier modeling methods like Booch, OMT, and OOSE. Adopted by the Object Management Group (OMG), UML standardizes system design and is widely used in software development.

3 UML Diagram :

UML diagrams are graphical tools used to visualize, design, and document the structure and behavior of a system, mainly in software development.

3.1 Class Diagram :

Shows the structure of a system by representing its classes, attributes, methods, and the relationships between them.

3.2 Use Case Diagram :

Describes the functional requirements of a system by showing the interactions between users (actors) and the system (use cases).

3.3 Sequence Diagram :

Illustrates the interaction between objects in a system over time, showing the flow of messages and the order of execution.

4 Analysis and Design :

In our conception, we used three types of diagrams:

4.1 Class Diagram :

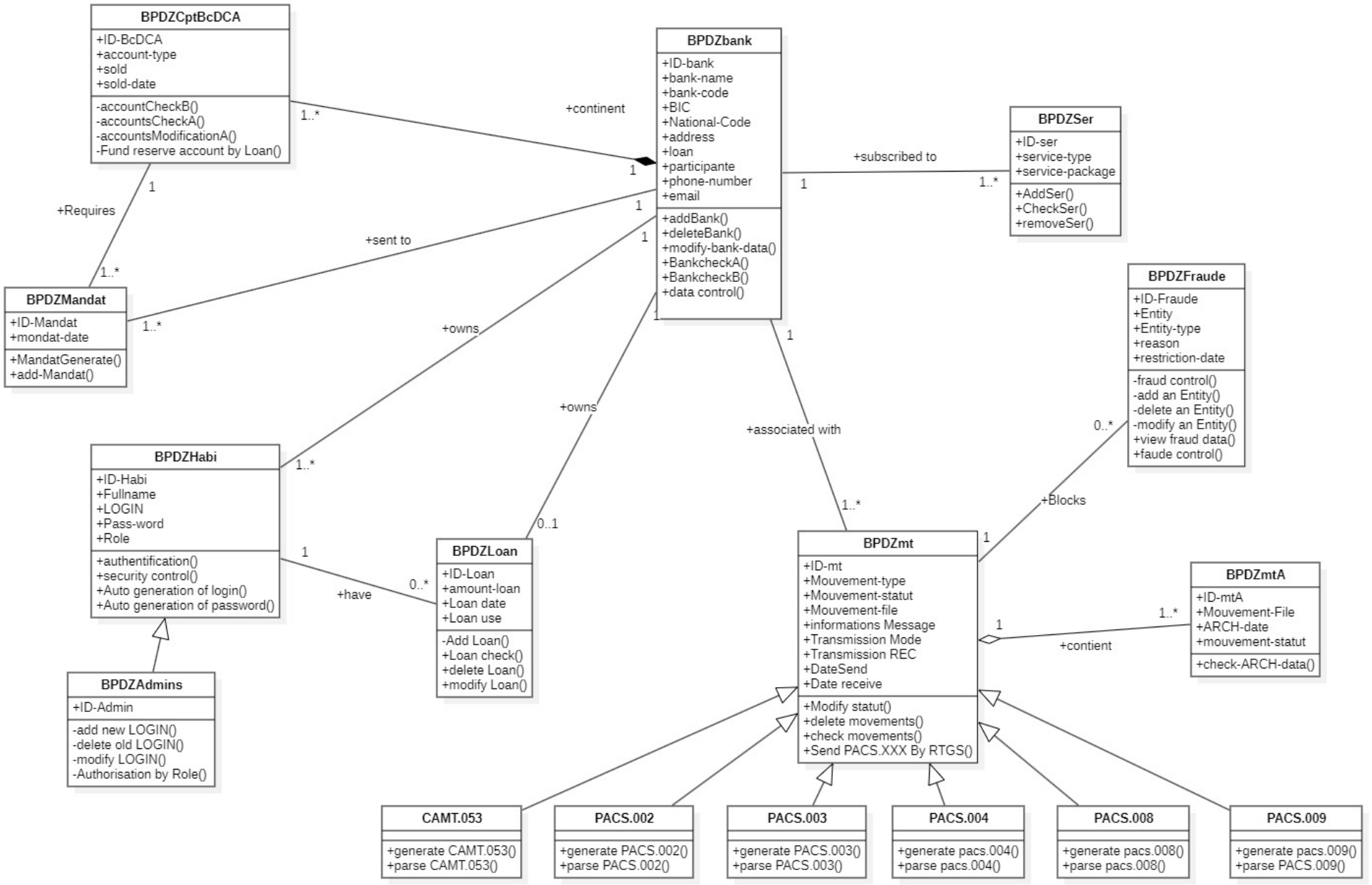


Figure 2.1: **Class Diagram**

This class diagram represents the core structure of the **BPDZ banking platform**, modeling key system entities, their attributes, operations, and relationships.

- **BPDZHabi** represents users of the system, supporting authentication, security control, and automatic credential generation.
- The **BPDZAdmins** class inherits from **BPDZHabi** and is responsible for managing user logins and authorizations based on roles.
- **BPDZbank** contains banking information, with operations to add, delete, and verify bank data.
- **BPDZCptBcDCA**, **BPDZMandat**,and **BPDZLoan** manage accounts, mandates, and loans respectively, each with specialized methods for control and updates.
- **BPDZSer** handles the management of bank services and packages.
- **BPDZFraude** is responsible for fraud control, including adding, modifying, and blocking suspicious entities.
- **BPDZmt**is the main class for handling payment and transaction movements, supporting SEPA message transmission via RTGS.
- **BPDZmtA** archives movement data for long-term storage and audit purposes.
- The system supports standard financial message formats, including **PACS.008**, **PACS.009**, **PACS.002**, **PACS.003**, **PACS.004**, and **CAMT.053**, with methods to generate and parse these messages.
- The diagram includes important relationships like **inheritance** (e.g., Admins inherit from users), **associations** between entities (e.g., banks own loans), and **cardinalities** that define how many instances relate to one another.
- **Cardinalities:** Clearly indicated (e.g., 1.., 0.., etc.) to manage dependencies and multiplicities.

4.2 Use Case Diagram :

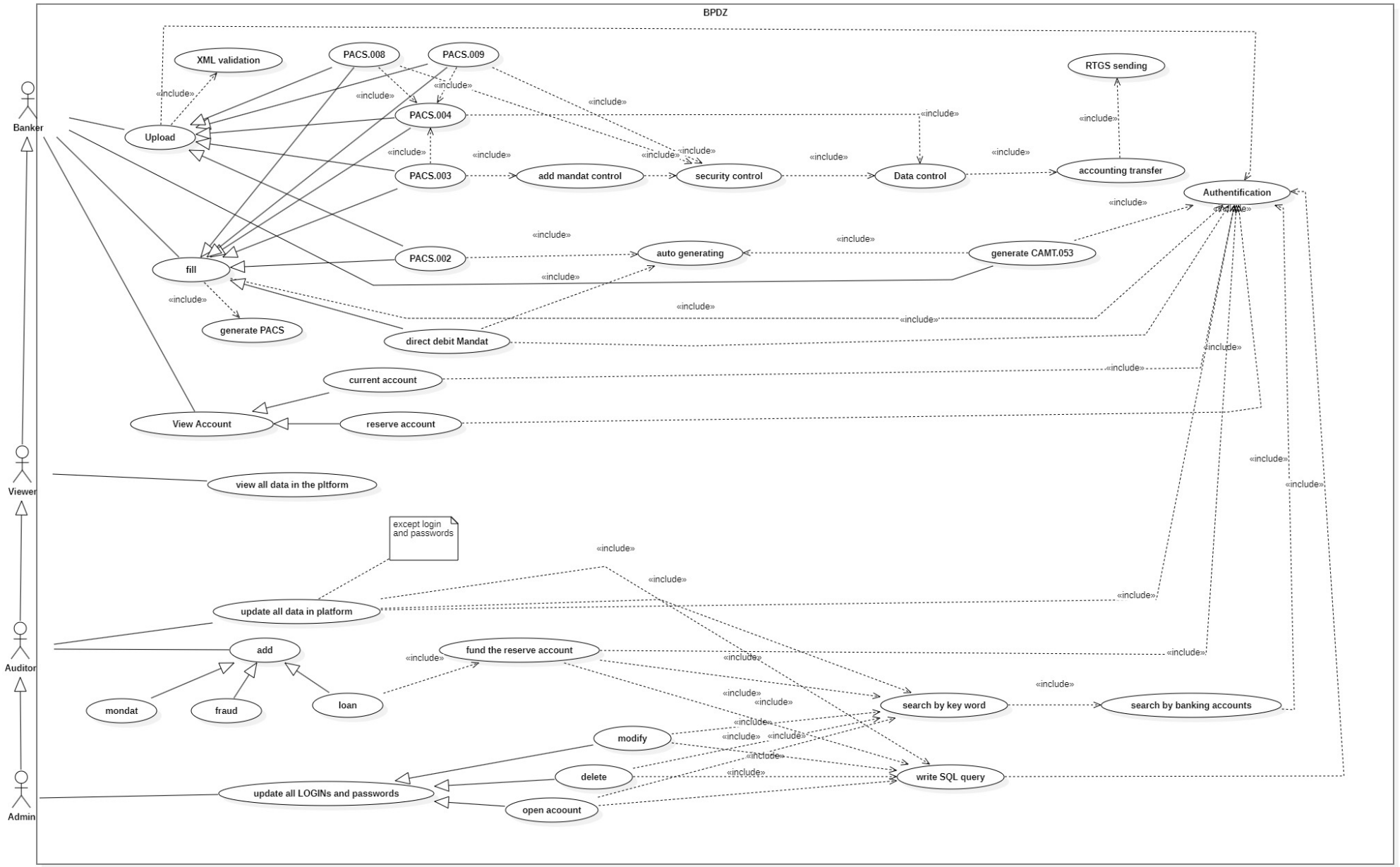


Figure 2.2: **Use Case Diagram**

A **Use Case Diagram** shows the **functionalities** (use cases) that a system provides and the **interactions** of **different user roles** (actors) with those functionalities. It helps stakeholders understand how the system is used. The BPDZ platform is designed to manage banking operations with clearly defined roles and responsibilities for different users. The primary actors in the system are as follows :

- **Banker**: Responsible for operational tasks such as uploading financial message files (e.g., PACS.008, PACS.009), filling transaction forms, generating PACS messages, and viewing account data (current and reserve accounts).
- **Viewer**: This actor has read-only access to platform data. Viewers can consult account information and view system-wide data without making any modifications.
- **Auditor**: In charge of overseeing data consistency and compliance. Auditors are authorized to update all platform data except login credentials. They can also manage loans, fund reserve accounts, and execute various search or analysis queries.
- **Administrator (Admin):** : Holds full system privileges. Administrators are responsible for opening, modifying, and deleting user accounts, as well as managing login credentials. They can also perform advanced data queries based on keywords or banking account references.

The **main functionalities (Use Cases)** supported by the platform include:

- Uploading financial files with integrated controls (e.g., XML validation, mandate checks).
- Filling in forms and generating PACS and CAMT.053 messages.
- Executing real-time fund transfers using RTGS.
- Managing loans and reserve accounts.
- Performing secure authentication for all users.
- Running advanced queries and account management actions.

The system uses two types of relationships:

- **include**:Indicates that a use case must always include another (mandatory behavior).
- **extend**: Represents optional or conditional actions that may be triggered depending on the context.
- **Inheritance relationship (generalization/specialization):** used when several actors share common behaviors.

4.3 Sequence Diagram :

4.3.1 PACS.002 : Payment Status Notification

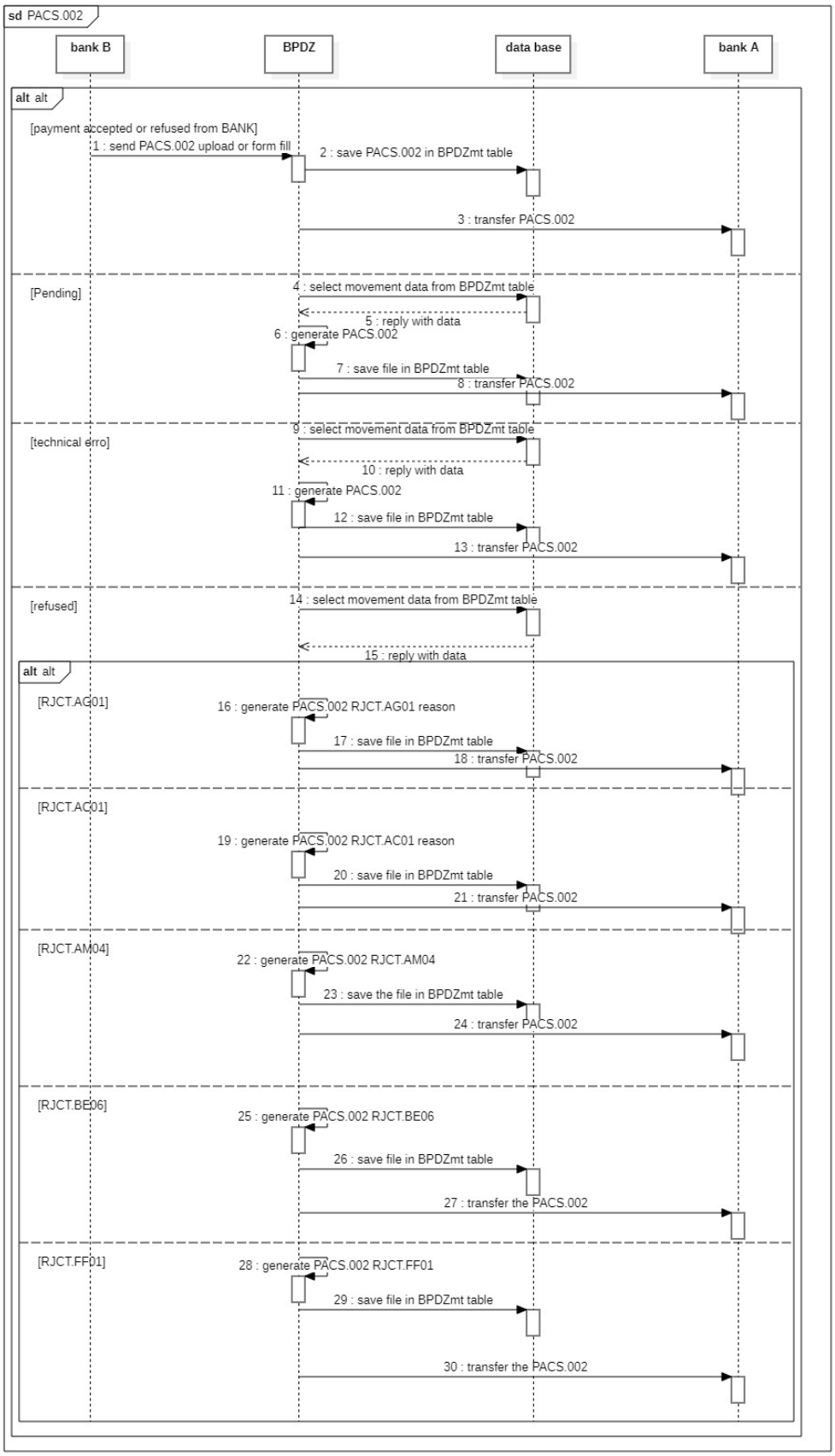


Figure 2.3: **Sequence Diagram of PACS.002**

This sequence diagram shows how a PACS.002 message (used to reject or respond to a payment) flows between **Bank B**, the **BPDZ platform**, the **database**, and **Bank A**. Depending on the situation (accepted, pending, technical error, or refused), BPDZ processes the request, generates the appropriate PACS.002 message, saves it, and sends it to Bank A. If the payment is refused, different rejection codes (like RJCT.AG01, RJCT.AC01, etc.) are used to explain the reason.

Table 2.4: **Detailed Scenarios of PACS.002**

Scenario	**Messages**	**Status/Reason**
Payment Accepted	$1 \rightarrow 4$	Accepted
Payment Pending	$5 \rightarrow 8$	Pending
Technical Error	$9 \rightarrow 13$	Technical Error
Payment Refused	$14 \rightarrow 15$	Refused
Rejection : AG01	$16 \rightarrow 18$	General rejection
Rejection : AC01	$19 \rightarrow 21$	Account closed
Rejection : AM04	$22 \rightarrow 24$	Invalid mandate
Rejection : BE06	$25 \rightarrow 27$	Format error
Rejection : FF01	$28 \rightarrow 30$	Fraud suspicion

4.3.2 PACS.003 : Direct Debit Request

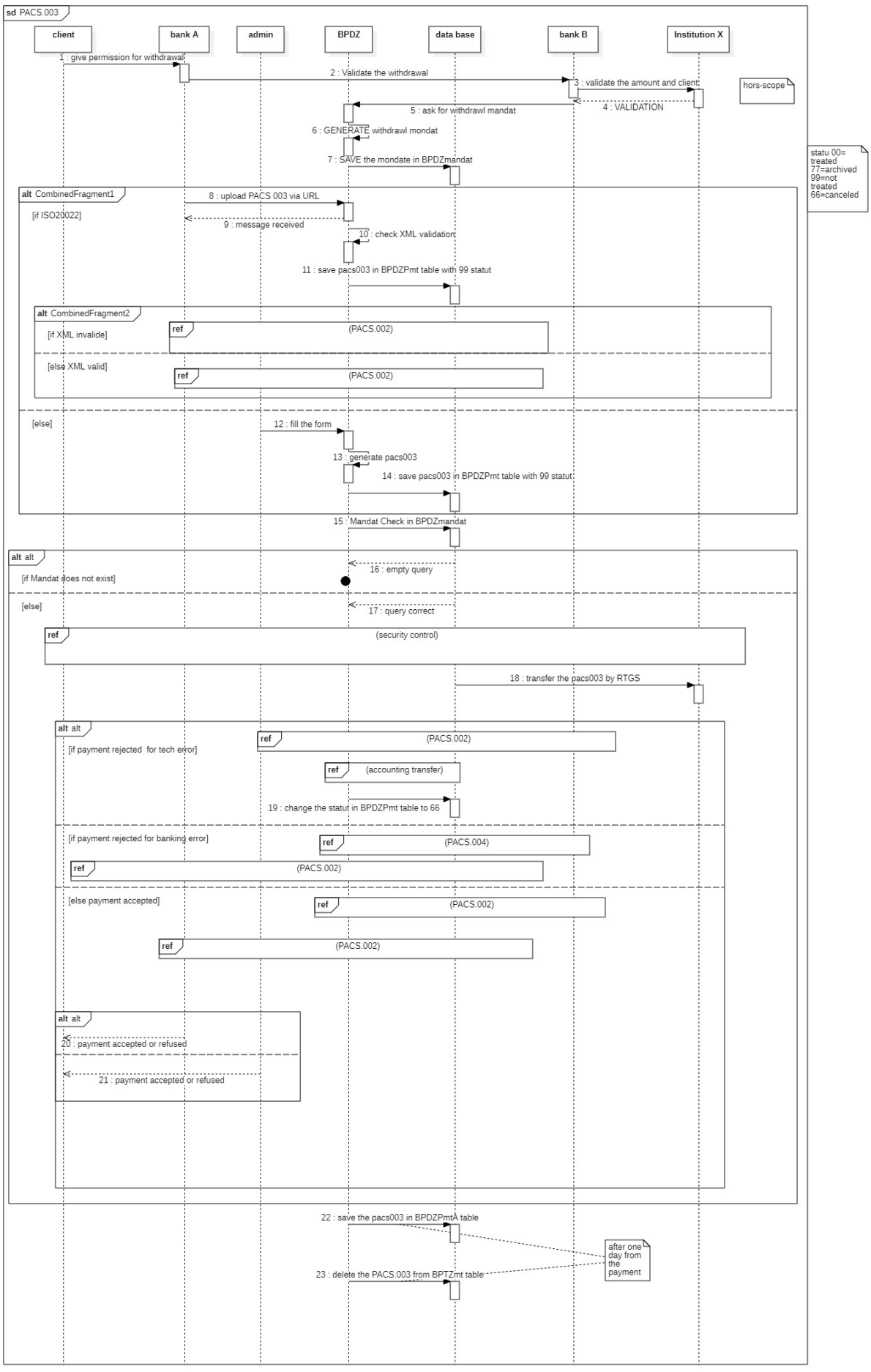


Figure 2.4: **Sequence Diagram of PACS.003 Message Processing**

The diagram sequence shows the flow of a PACS.003 payment request from initiation to completion. It includes validation, mandate creation, message upload, and transfer to Bank B. The process ends with acceptance, rejection, or error handling, followed by client notification and archiving.

Table 2.5: **Detailed Scenarios of PACS.003**

Scenario	**Messages**	**Status/Reason**	**Explanation**
Authorization & Validation	1 → 4	Accepted (initial phase)	Client authorizes; validation by Bank A, Admin, Institution X.
Mandate Request & Creation	5 → 7	Pending	Mandate requested, generated, and saved in BPDZ-mandat.
PACS.003 Upload & XML Validation	8 → 11	Waiting for validation	PACS.003 uploaded; XML validated and saved with status 99.
Manual Form Entry (Fallback)	12 → 14	Manual entry	Used if XML is invalid; PACS.003 manually generated and saved.
Mandate Verification	15 → 17	Under verification	System checks for existing mandate; security control.
Payment Transfer	18	Transfer initiated	PACS.003 is sent to Bank B via RTGS.
Rejection Technical or Bank Error	19	Refused (status 66)	Payment rejected due to technical or banking error.
Payment Accepted	After 18	Accepted	Bank B accepts PACS.002 sent to confirm.
Notification of Status	20 → 21	Payment response	Client is informed if payment was accepted or refused.
Post-Payment Archiving	22 → 23	Archived	PACS.003 is saved in the archive and deleted from the temporary table after one day.

4.3.3 PACS.004: Return Funds

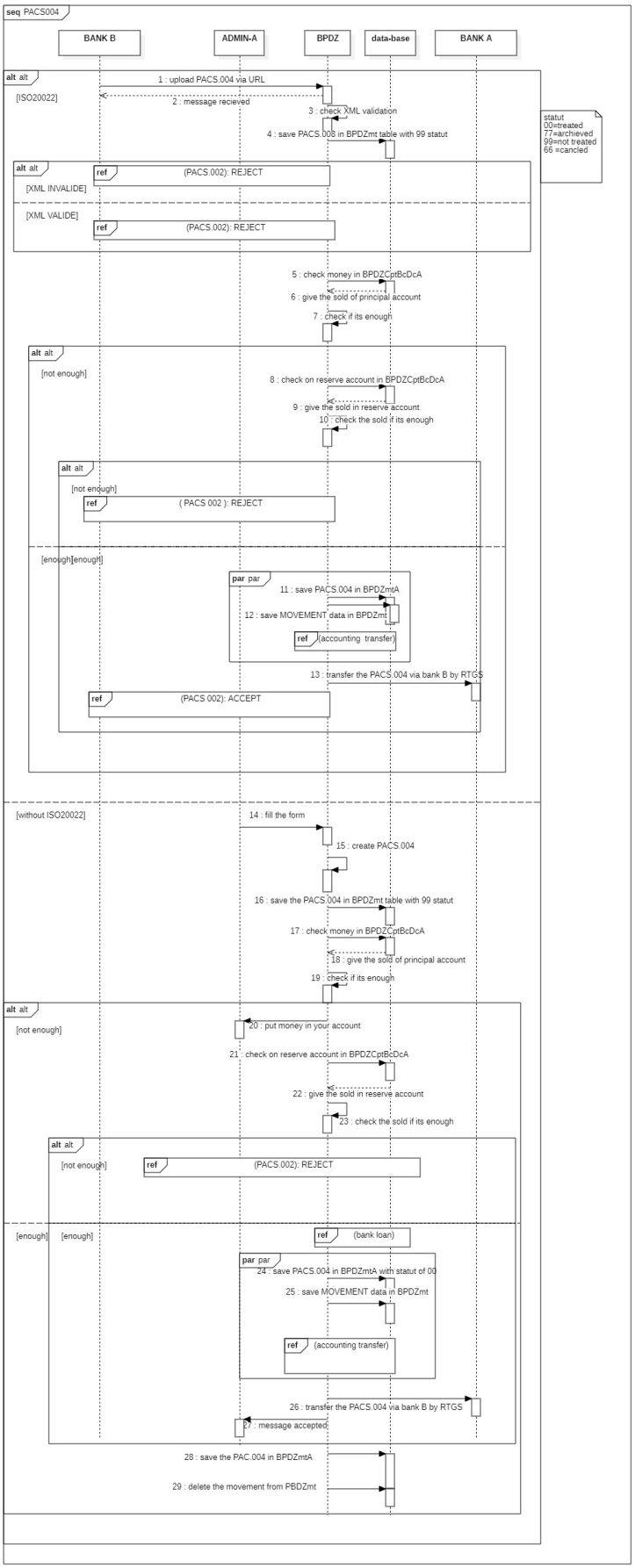


Figure 2.5: **Sequence Diagram of PACS.004**

This sequence diagram represents the flow of the PACS.004 message, typically used to return funds or report payment reversals. It includes XML validation, account balance checks, optional fallback to reserve accounts, and rejection or acceptance of the payment. The process accommodates both ISO20022-based flows and manual inputs, ending with transaction confirmation and archiving.

Table 2.6: **Detailed Scenarios of PACS.004**

Scenario	**Messages**	**Status/Reason**	**Explanation**
PACS.004 Upload & XML Validation	1 → 4	Processing	Bank B uploads PACS.004, XML is validated and saved with status 99.
Invalid XML Rejection	ref (alt XML_INVALID)	Rejected	PACS.004 rejected due to invalid structure.
Valid XML → Fund Check	5 → 7	Checking Balance	BPDZ checks principal account balance.
Use Reserve Account (if needed)	8 → 10	Reserve Account Checked	If principal funds are not enough, reserve account is evaluated.
Rejection for Insufficient Funds	ref (alt not enough)	Rejected	Both accounts have insufficient funds → PACS.002 sent to reject.
Acceptance → Fund Movement & Transfer	11 → 13	Accepted	Enough funds found → transfer PACS.004 and save transaction.
Manual PACS.004 Form Creation	14 → 16	Manual Entry	Non-ISO uploads trigger manual PACS.004 generation and save.
Manual Fund Checking	17 → 19	Checking Balance	Principal account checked for sufficient balance.
Reserve Fallback (Manual Path)	20 → 23	Reserve Check (Manual)	Reserve checked if principal balance is insufficient.
Rejection (Manual Insufficiency)	ref (alt not enough)	Rejected	Same as above, if funds not enough → PACS.002 reject sent.
Acceptance via Bank Loan	24 → 26	Accepted (Loan-backed)	Sufficient funds or bank loan → PACS.004 transferred via RTGS.
Message Acknowledgment	27	Message Accepted	Bank B confirms message accepted.
Final Archiving & Cleanup	28 → 29	Archived	PACS.004 saved in archive and movement deleted from processing table.

4.3.4 PACS.008 : customer credit transfers

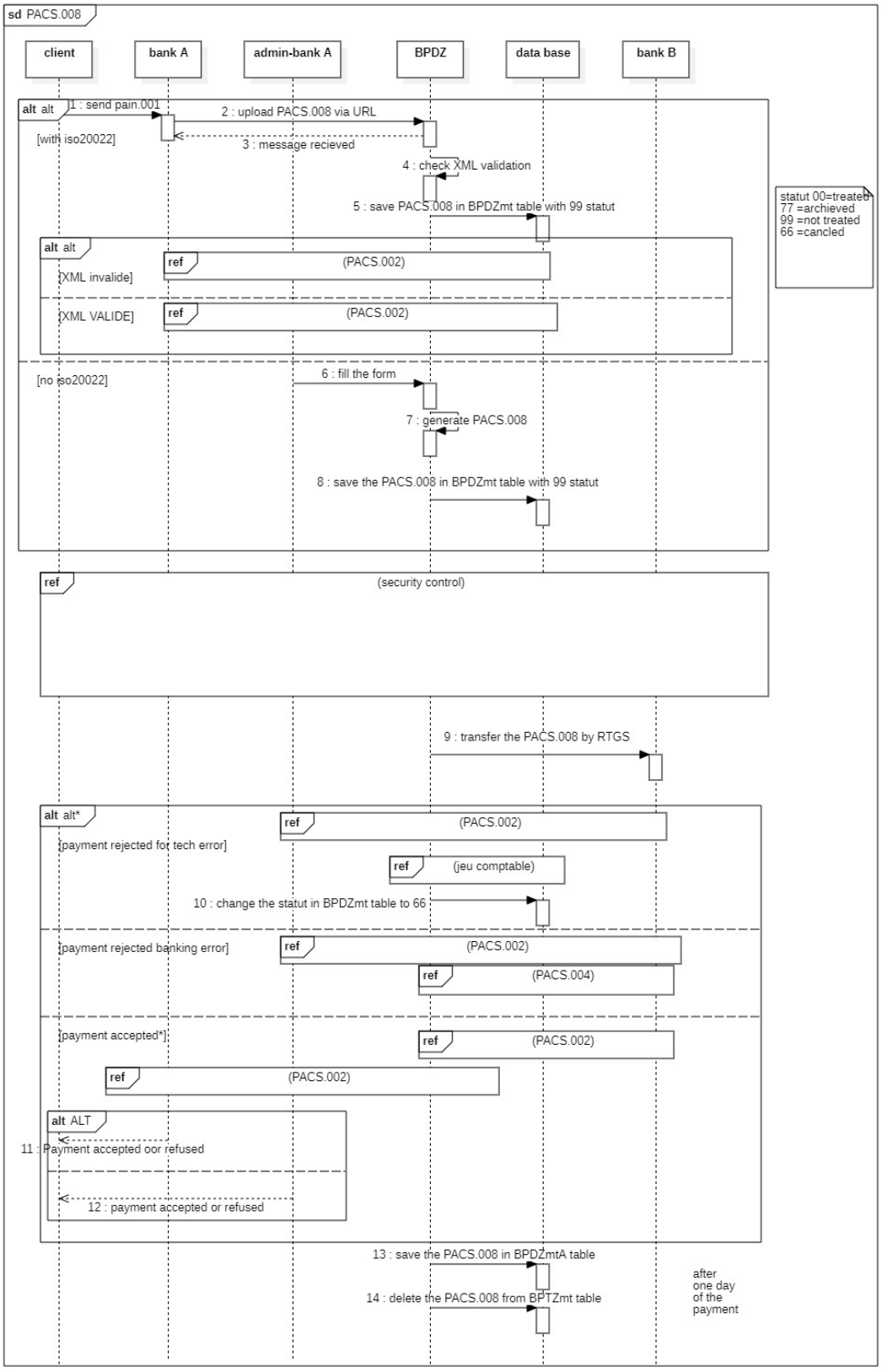


Figure 2.6: **Sequence Diagram of PACS.008**

This sequence diagram shows how a **PACS.008 message** (used for **customer credit transfers**) is processed between a **client**, **Bank A**, **Admin Bank A**, the **BPDZ platform**, the **database**, and **Bank B**. It handles different flows based on whether the transfer is sent as an ISO20022 XML file or filled manually. The system checks XML validity, performs security controls, transfers the message using RTGS, and sends PACS.002 rejection messages if needed. Finally, the PACS.008 is archived or deleted after processing.

Table 2.7: **Detailed Scenarios of PACS.008**

Scenario	Messages	Status/Reason	Explanation
PACS.008 Upload & XML Validation	1 → 5	Processing	Client sends pain.001. Bank A uploads PACS.008 via URL. BPDZ validates the XML and saves it with status 99 in the BPDZmt table.
Invalid XML Rejection	ref (alt XML invalid)	Rejected	XML is invalid → PACS.002 is generated and payment is stopped.
Valid XML → Continue	ref (alt XML valid)	Processing	XML is valid → PACS.002 is generated and process continues.
Manual Form Creation (no ISO20022)	6 → 8	Manual Entry	If no ISO20022 file, admin fills a form manually. PACS.008 is generated and saved in BPDZmt with status 99.
Security Check & Transfer via RTGS	9	Processing	After internal security control, PACS.008 is transferred to Bank B via RTGS.
Technical Error Rejection	ref (alt tech error)	Rejected	Payment rejected due to technical issue → PACS.002 sent. Status updated to 66 in BPDZmt.
Banking Error Rejection	ref (alt banking error)	Rejected	Payment rejected due to banking issue → PACS.004 sent.
Successful Payment Acceptance	ref (alt accepted)	Accepted	Payment is accepted by Bank B → PACS.002 sent confirming success.
Confirmation to Admin Bank A	11 → 12	Accepted / Rejected	Admin Bank A receives final status: either accepted or refused.
Archiving the Transaction	13	Archived	Accepted PACS.008 is saved in the BPDZmtA archive table.
Cleanup – Remove Old Entry	14	Deleted	One day after payment → PACS.008 entry is deleted from BPDZmt.

4.3.5 PACS.009 : credit transfers between banks

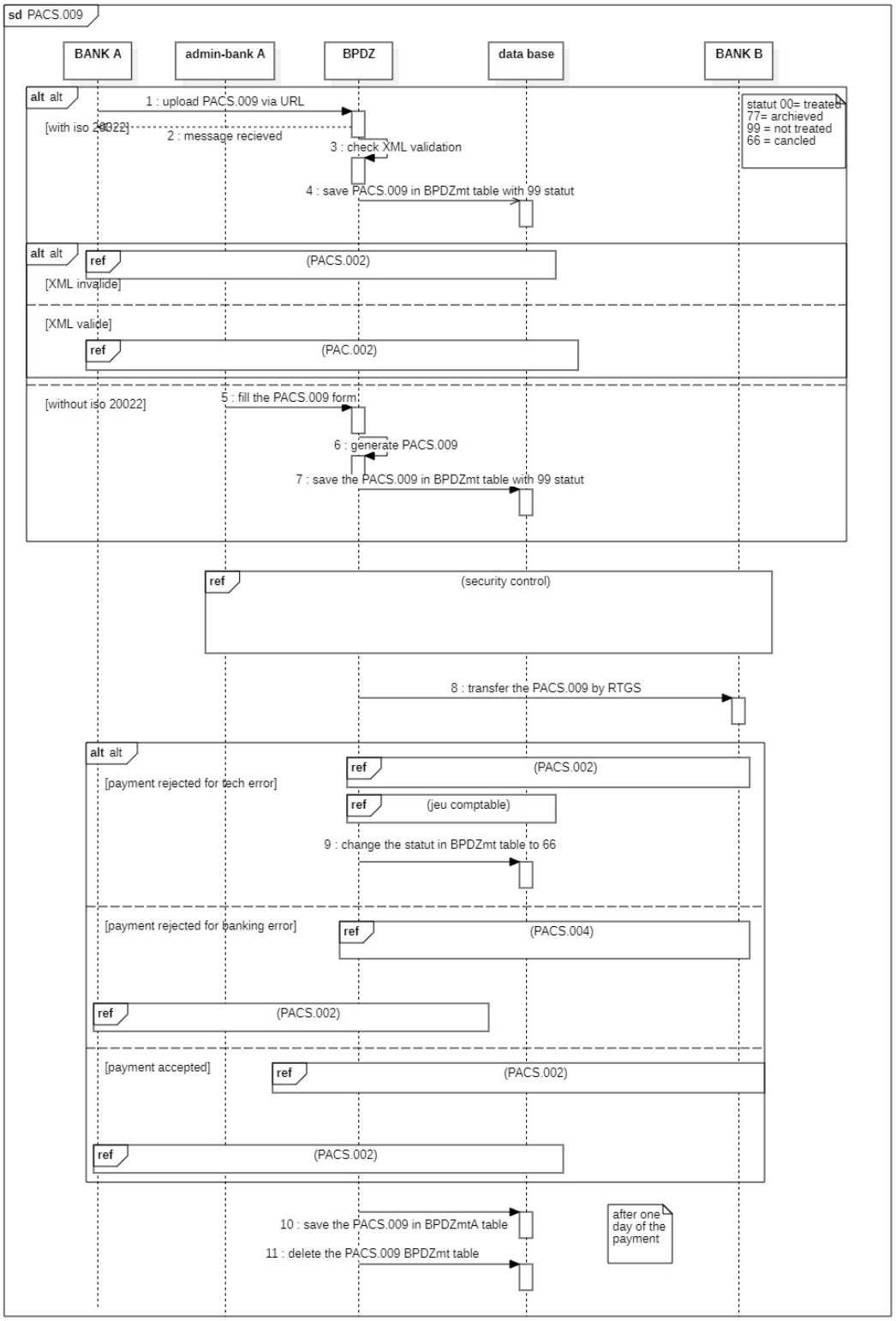


Figure 2.7: **Sequence Diagram of PACS.009**

This sequence diagram shows **PACS.009 message** (used for credit transfers between banks).

NOTE: PACS.009 same as PACS.008 the only difference this one is used for credit transfers between banks while PACS.008 used for costumer credit transfer.

4.3.6 Camt 053 : Bank Statement

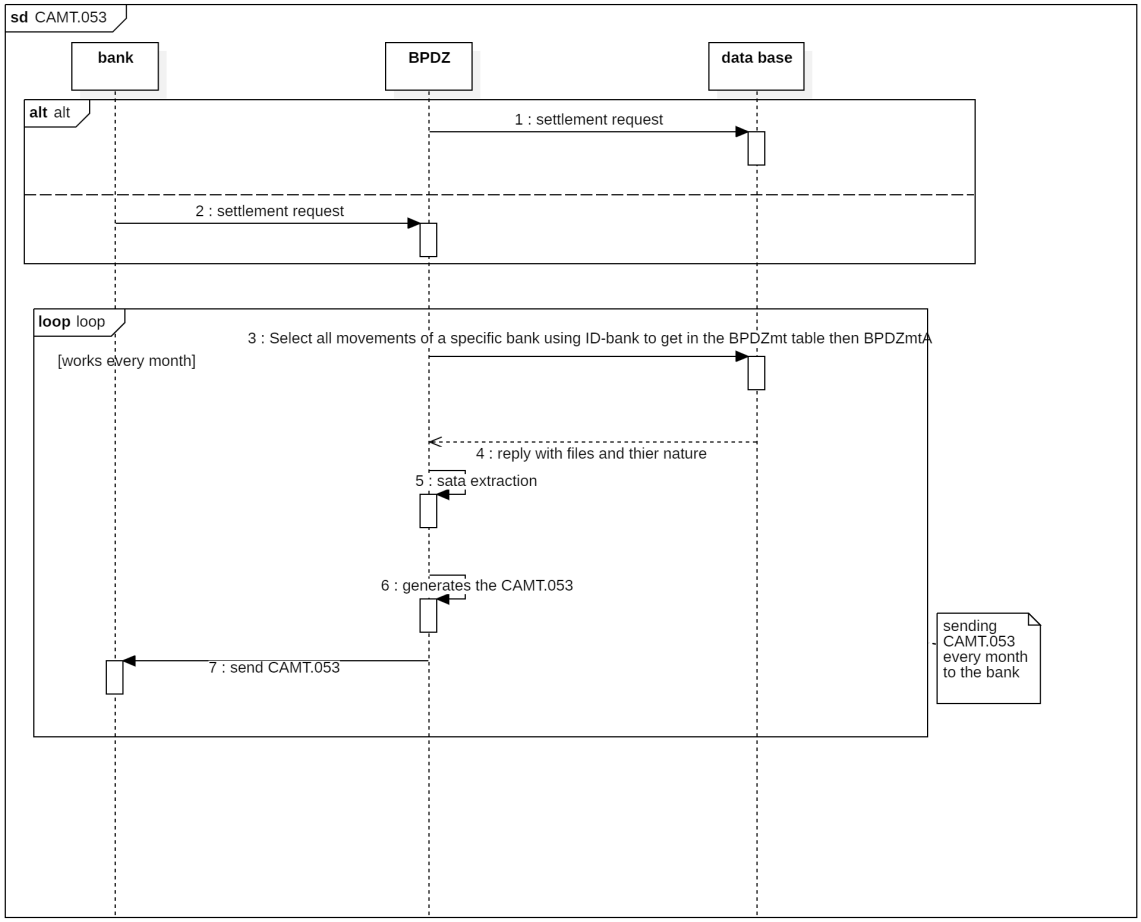


Figure 2.8: **Sequence Diagram of Camt 053**

This sequence diagram represents the **monthly CAMT.053 generation and sending process** in the BPDZ platform.

Table 2.8: **Detailed Scenarios of Camt 053**

Step	Description
1-2	The bank initiates a settlement request to the BPDZ system.
3	Every month, BPDZ selects all account movements of that specific bank using its ID from the BPDZmt and BPDZmtA tables.
4	The database replies with the list of transactions and their nature (credits/debits).
5	BPDZ performs data extraction from the received files.
6	BPDZ generates the CAMT.053 message using the extracted data.
7	The CAMT.053 statement is sent to the bank.

NOTE: This process runs monthly.

4.3.7 Security control :

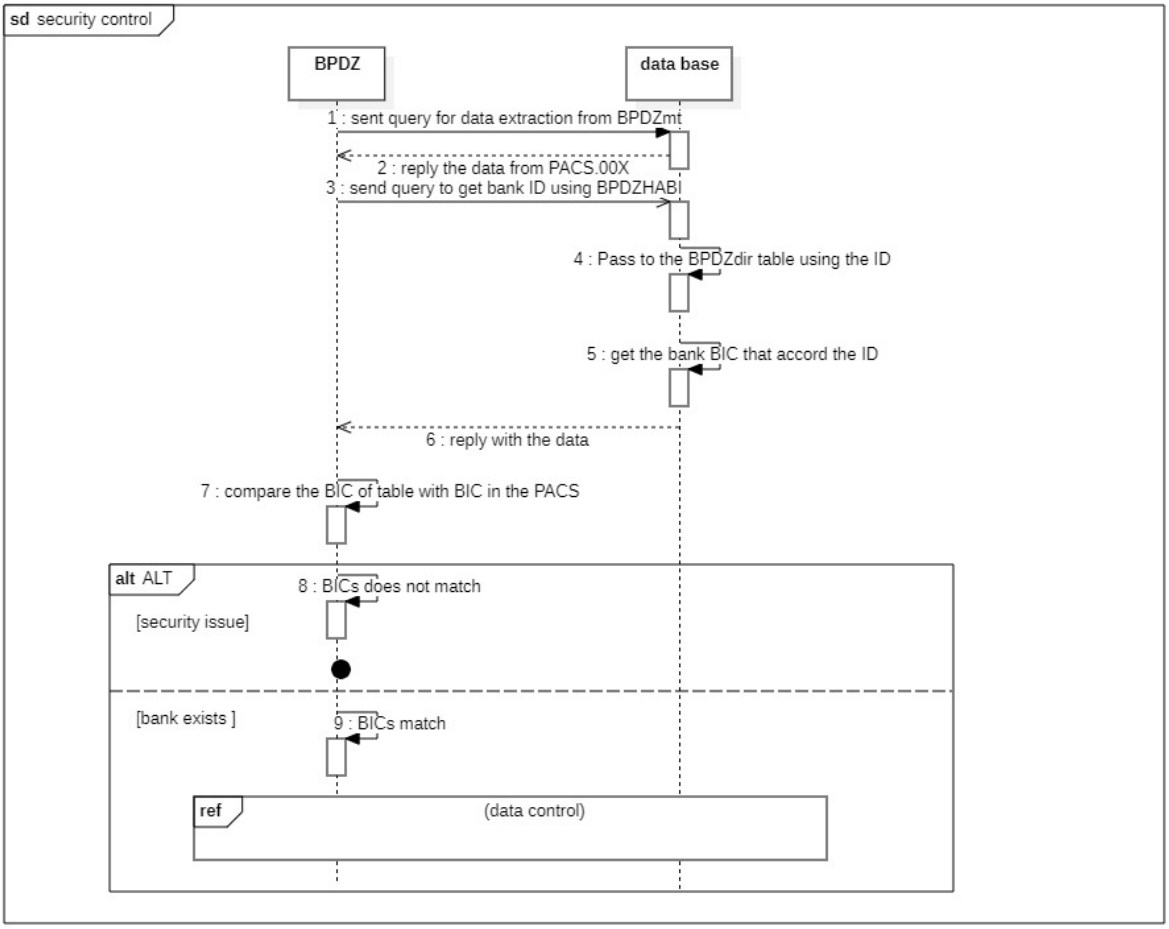


Figure 2.9: **Sequence Diagram of Security control**

This sequence ensures the **validity and authenticity** of a PACS message by comparing the BIC (Bank Identifier Code) from the message to the one registered in the system.

Table 2.9: **Detailed Scenarios of Security control**

Step	Action	Explanation
1	Send query to extract data	BPDZ starts by extracting transaction info from BPDZmt.
2	Receive PACS data	Database replies with PACS.00X transaction data.
3	Request bank ID	BPDZ queries bank ID using the BPDZHabi table.
4	Search directory	Data is passed to BPDZdir to match the bank by ID.
5	Get BIC	Database retrieves the BIC (Bank Identifier Code) for the ID.
6	Return BIC data	BIC is returned to BPDZ.
7	Compare BICs	Compare BIC from database with BIC in the PACS message.
8	BICs do not match	Security issue detected invalid or tampered message.
9	BICs match	Message is valid and the bank exists.
→	Reference to next step	If BICs match, continue with data control (validation process).

4.3.8 Data Control :

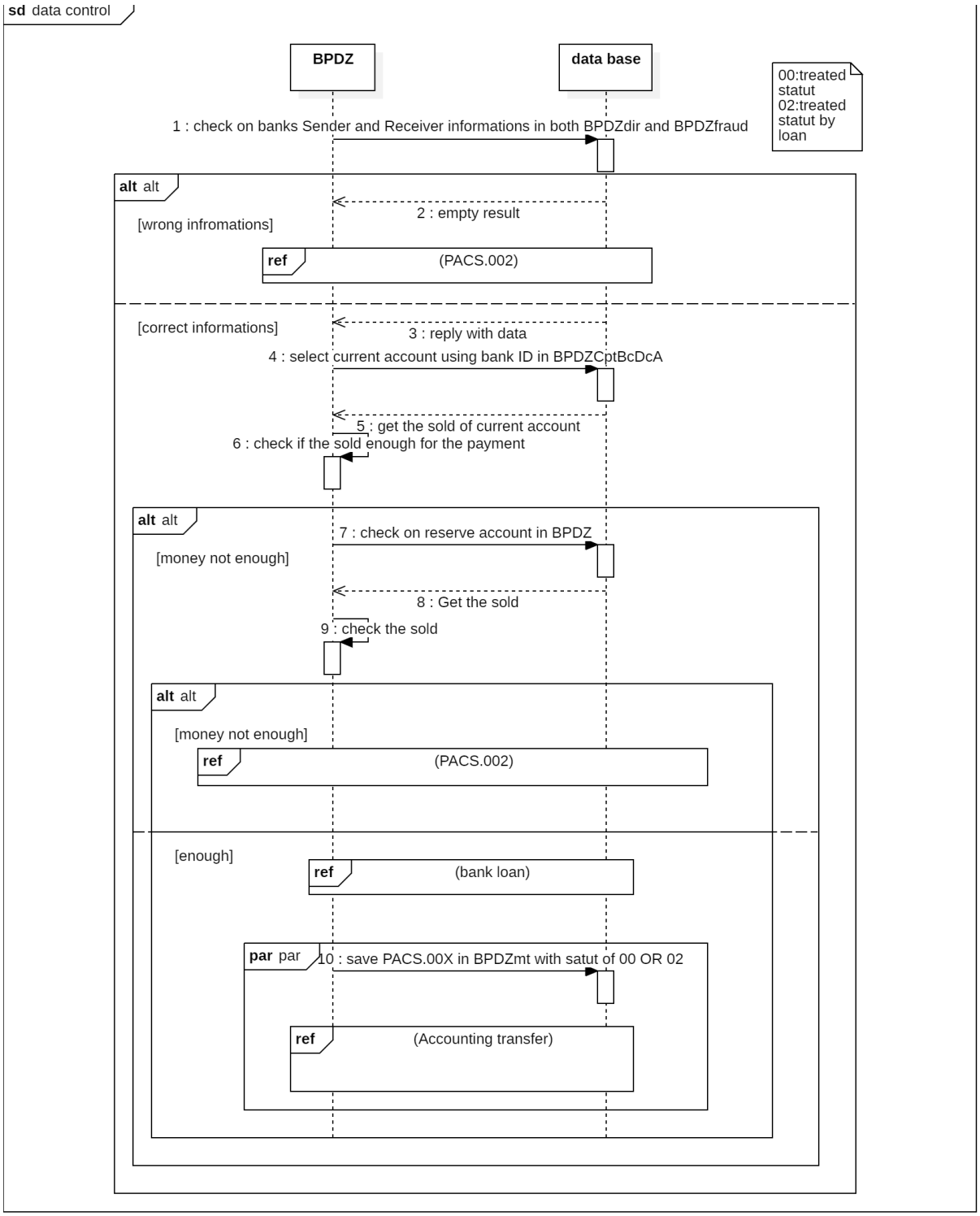


Figure 2.10: **Sequence Diagram of Data Control**

This sequence diagram explains how the system verifies sender and receiver bank information, checks account balances, and decides whether a payment can proceed.

Table 2.10: **Detailed Scenarios of Data Control**

Scenario	**Messages**	**Status/Reason**	**Explanation**
Check Bank Information	$1 \rightarrow 2$	Validation	Verifies bank sender/receiver info in BPDZdir and BPDZfraud.
Wrong Information $\rightarrow$ Rejection	ref (wrong info)	Rejected (PACS.002)	If wrong bank data, payment is immediately rejected.
Correct Information $\rightarrow$ Balance Check	$3 \rightarrow 6$	Checking Balance	Account selected, balance retrieved and checked.
Principal Account Insufficient	$7 \rightarrow 9$	Reserve Account Check	If not enough funds, check reserve account balance.
Insufficient Funds Overall	ref (not enough)	Rejected (PACS.002)	Both principal and reserve have insufficient funds.
Enough Funds or Loan Taken	ref (enough)	Accepted (Bank Loan)	Either sufficient funds found or loan accepted.
Save and Transfer Payment	10	Saved and Transferred	Transaction recorded in database and accounting transfer triggered.

4.3.9 Bank loan :

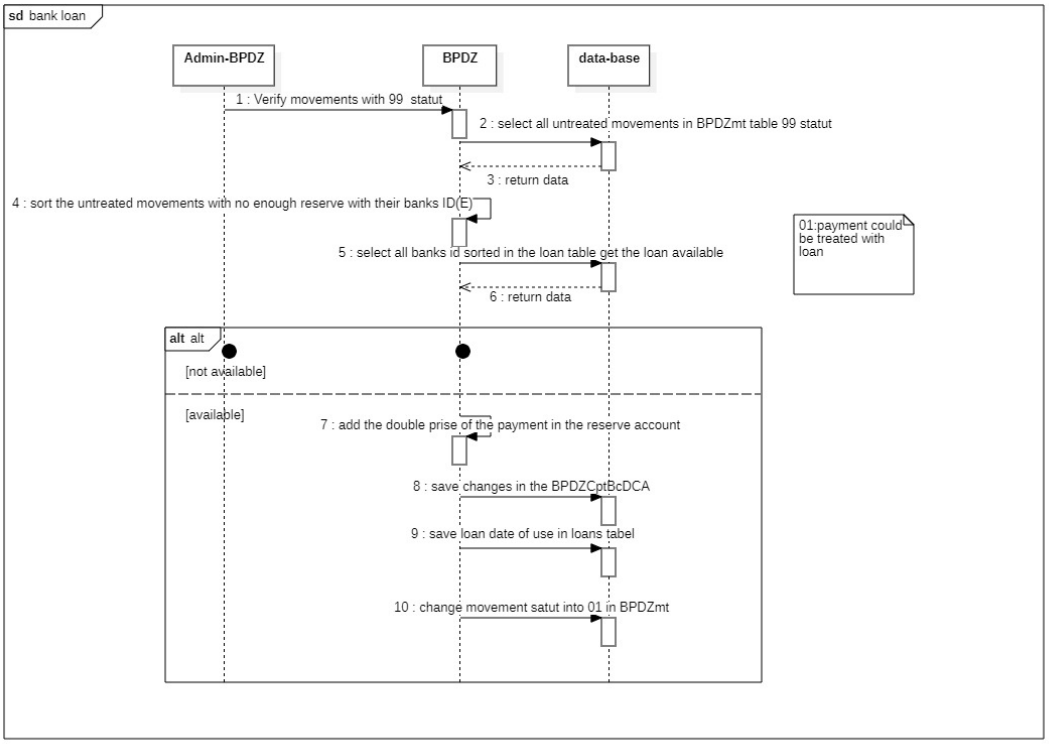


Figure 2.11: **Sequence Diagram of Bank loan**

This diagram shows how BPDZ verifies pending payment movements and checks if banks are eligible for automatic loans.

Table 2.11: **Detailed Scenarios of Bank loan**

Scenario	**Messages**	**Status/Reason**	**Explanation**
Verify Untreated Movements	$1 \rightarrow 3$	Validation	BPDZ checks untreated movements with status 01.
Check Auto Loan Availability	$4 \rightarrow 5$	Loan Verification	System checks if the bank has an auto loan facility.
No Auto Loan Available	alt (no loan)	No action	No loan, no changes made.
Auto Loan Available	alt (loan)	Loan Granted	Double the payment is reserved in the bank's reserve account.
Update Reserve and Loan Records	$6 \rightarrow 8$	Reserve & Loan Updated	Save reserve change and log loan usage data.

4.3.10 Accounting Transfer :

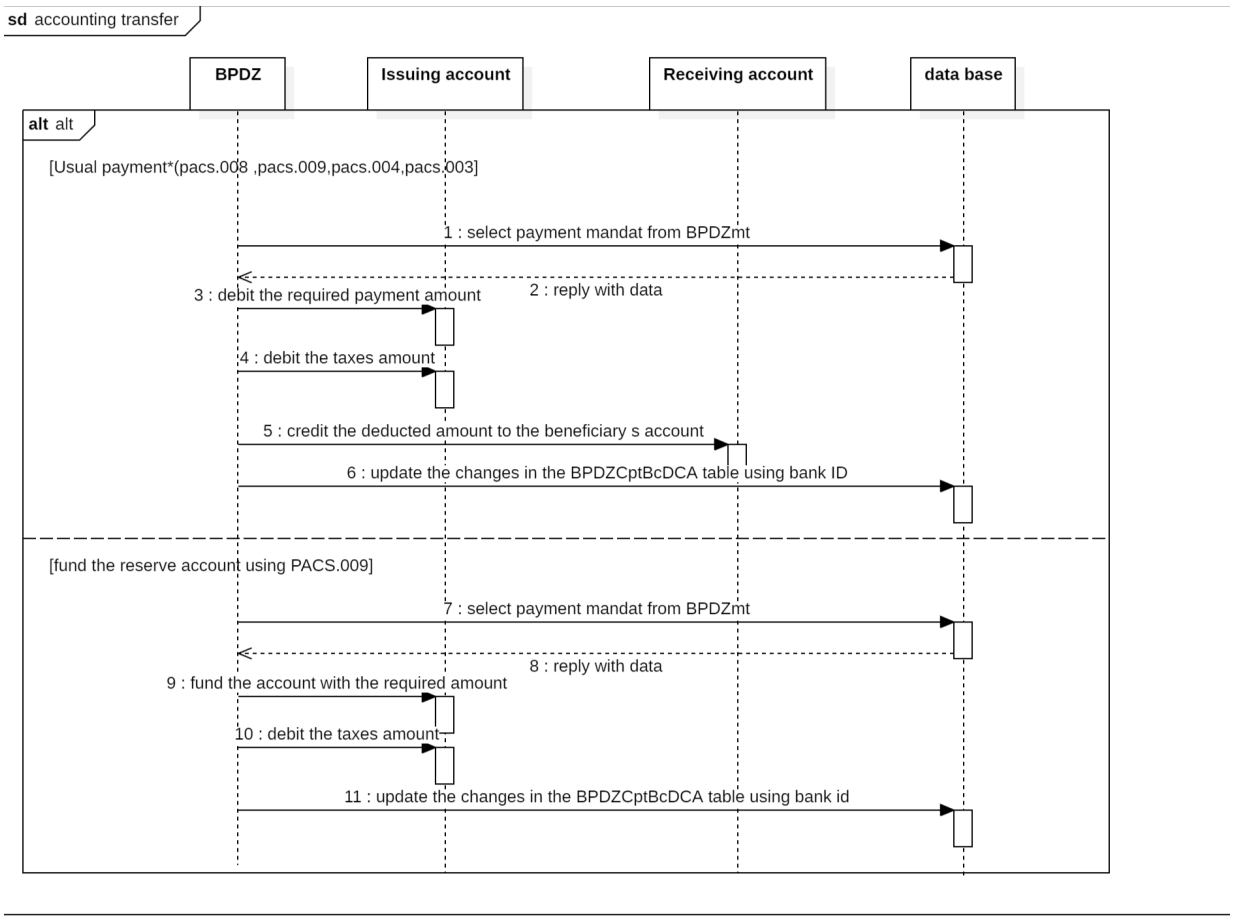


Figure 2.12: **Sequence Diagram of Accounting Transfer**

This sequence describes two cases: a usual payment and a reserve account funding using PACS messages. In each case, the system selects the payment mandate, debits the issuing account (including taxes), credits the receiving account.

Table 2.12: **Detailed Scenarios of Accounting Transfer**

Scenario	**Messages**	**Status/Reason**	**Explanation**
Usual Payment (PACS.008/009/004/003)	$1 \rightarrow 6$	Standard Transfer	Payment amount and taxes are debited; net amount is credited to beneficiary
Fund Reserve Account (PACS.009)	$7 \rightarrow 11$	Reserve Funding	Reserve account is funded similarly, with corresponding updates

4.3.11 RTGS (Real Time Gross Settlement) :

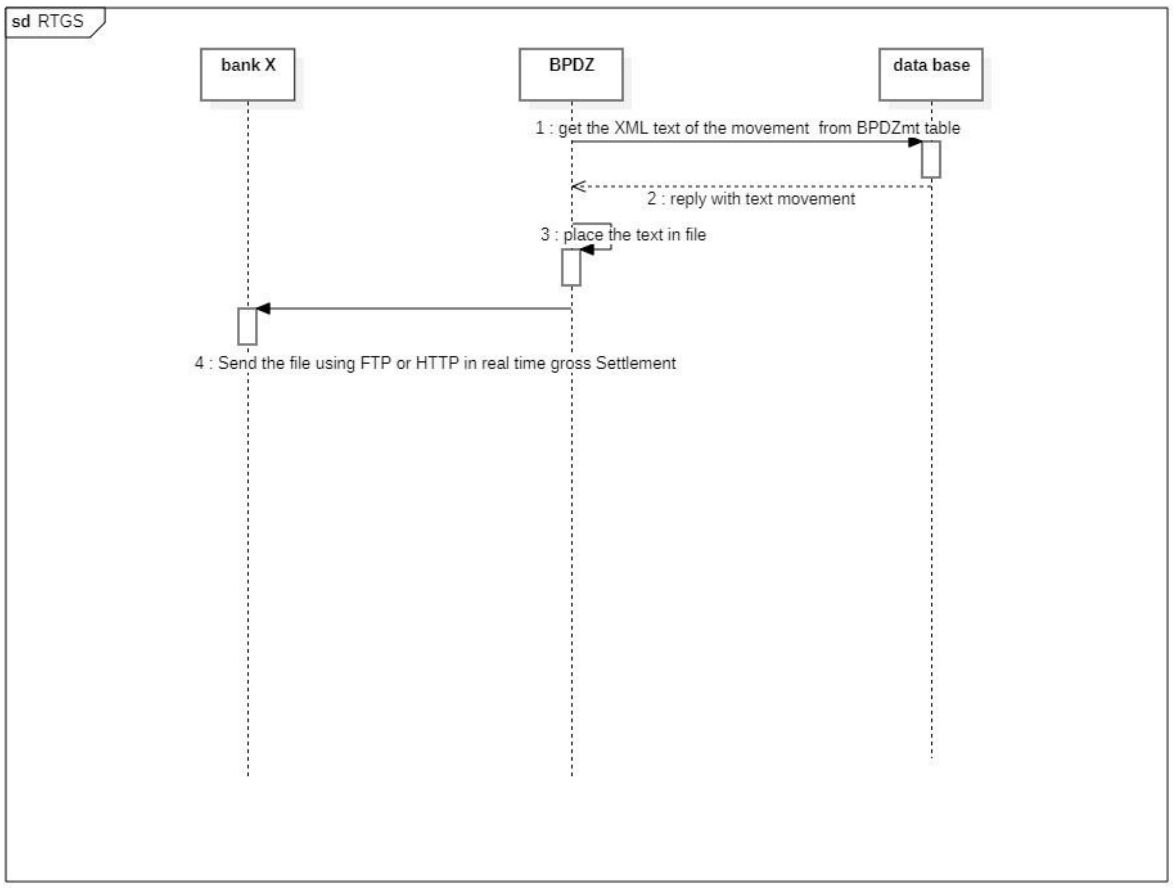


Figure 2.13: **Sequence Diagram of RTGS (Real Time Gross Settlement)**

4.3.12 Authentification :

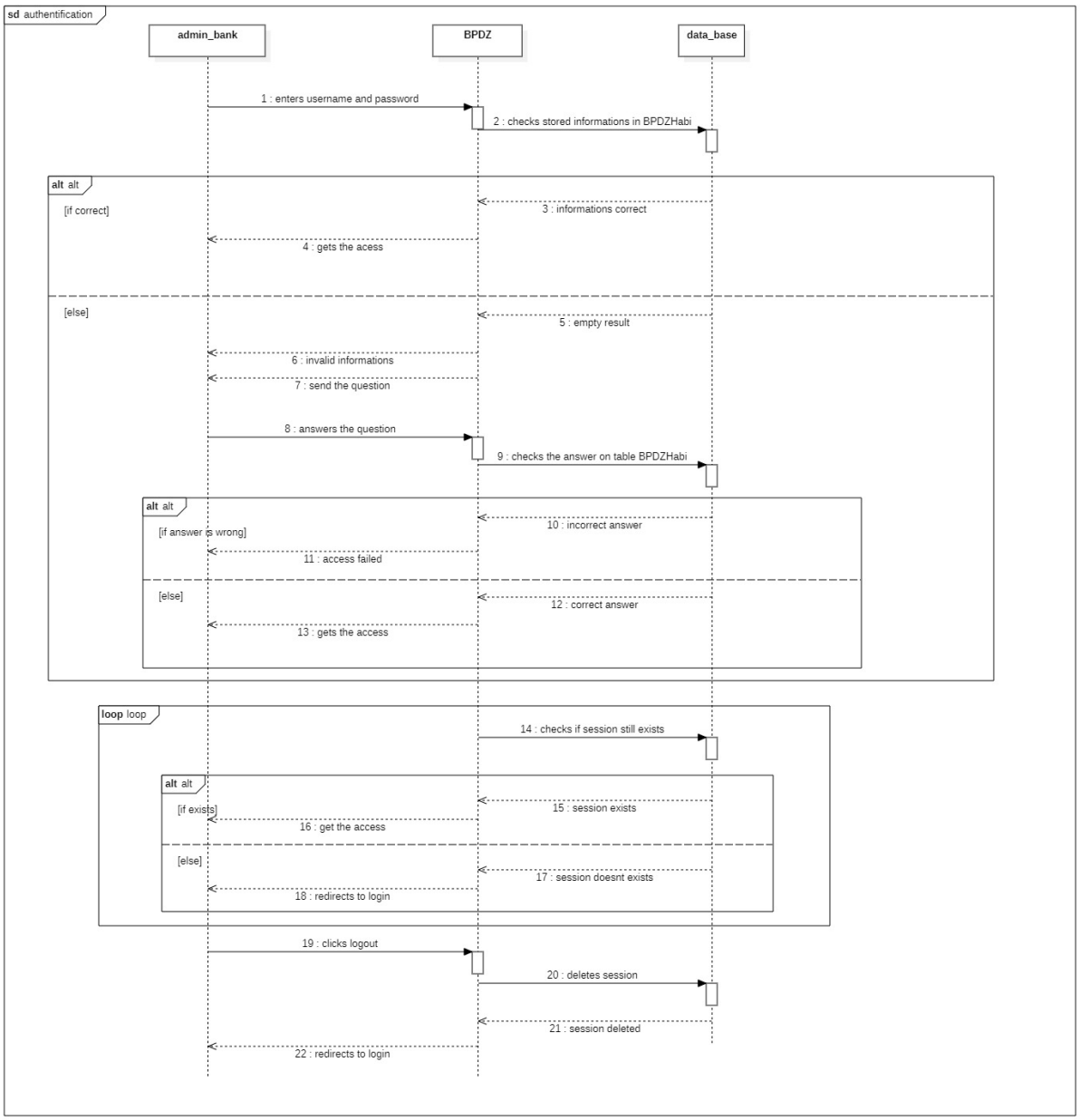


Figure 2.14: **Sequence Diagram of Authentification**

The sequence diagram describes the authentication process where the admin enters credentials, which are verified by BPDZ against the database. If correct, access is granted; otherwise, a security question is used for verification. The system also checks session validity and handles logout by deleting the session and redirecting to the login page.

4.3.13 Open and Delete Account :

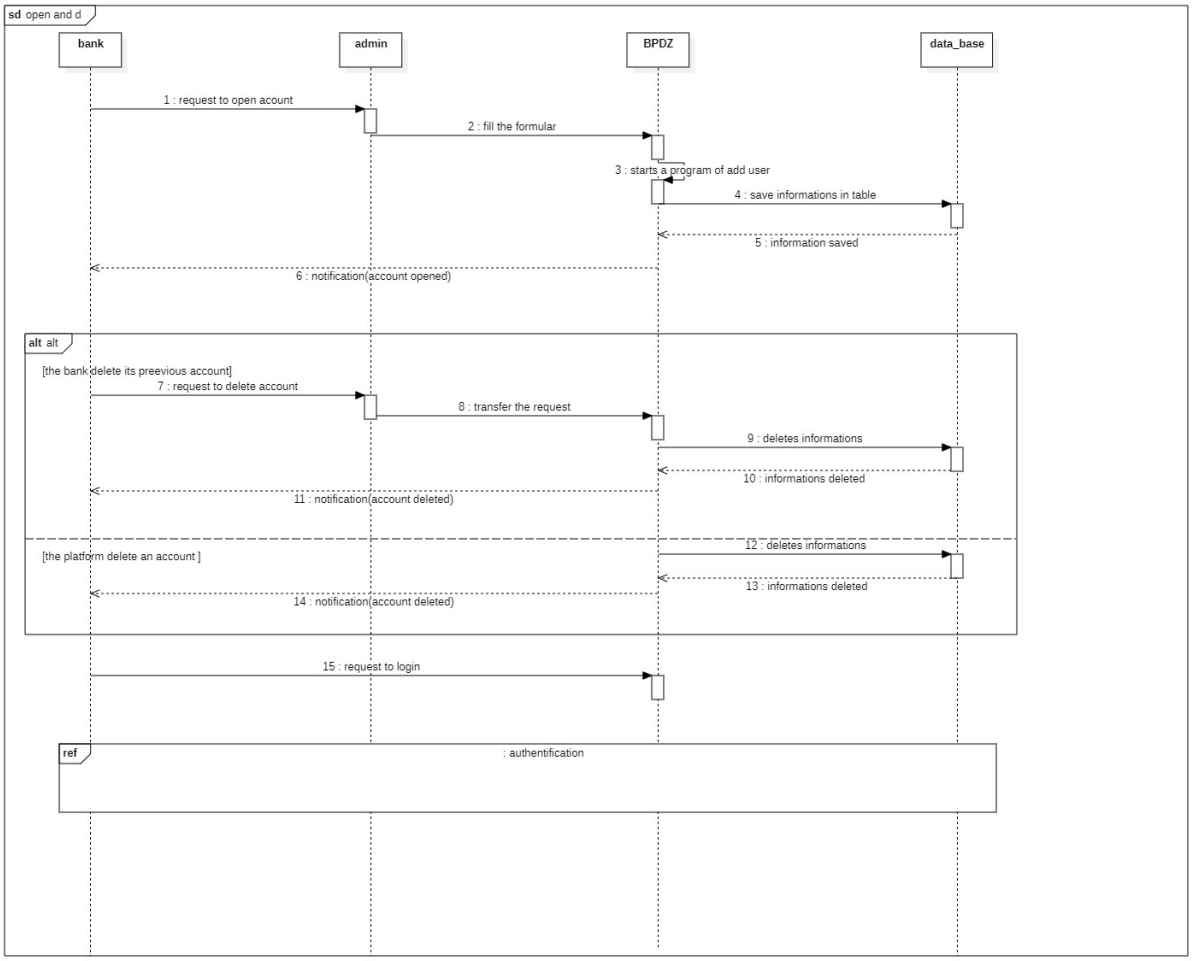


Figure 2.15: **Sequence Diagram of Open & Delete Account**

The sequence diagram shows how a bank opens or deletes an account. The bank asks to open an account, and the admin saves the details in the system. If the bank or platform wants to delete an account, the system removes the information and sends a message that the account was deleted. After that, it moves to the login process.

5 Conclusion

This chapter presented a comprehensive modeling of the BPDZ platform through detailed sequence diagrams and their corresponding scenarios. Each diagram illustrated a key functional aspect such as security control, data verification, loan processing, and authentication highlighting how the system manages interbank transactions in a secure, efficient, and automated manner. These visual and scenario based representations provide a clear roadmap for implementing a centralized interbank solution tailored to Algeria's financial modernization needs.

CHAPTER 3 : Implementation and Evaluation

1 Introduction :

This chapter outlines the implementation of our banking application, built within an interbank system modeled on ISO 20022 standards. We detail the technologies used Spring Boot and PostgreSQL and explain our unified table model for movement messages, designed to streamline integration and simplify maintenance.

2 Developement Environment

2.1 Tools Used

Tool	Version	Description
Spring Boot	3.x	Modern Java backend framework based on Spring
PostgreSQL	17	Relational database management system
Visual Studio Code	1.8x	Lightweight and extensible code editor
Eclipse IDE	2023-06	Robust integrated development environment
Maven	3.9	Java build and dependency management tool
Postman	-	Tool for testing RESTful APIs
Webhook	-	Automated event-triggered HTTP callbacks between apps.

2.2 Technical Choices

We chose **Spring Boot** due to its seamless integration with databases advanced dependency management support for RESTful APIs and compatibility with XML formats required by ISO 20022 messages. **PostgreSQL** was selected for its performance ACID compliance, and robustness.

For the front end development of our platform, we utilized **HTML, CSS**, and **JavaScript** to build a responsive and interactive user interface. These technologies allowed us to create structured layouts (HTML), apply consistent styling and branding (CSS), and implement dynamic client-side interactions (JavaScript), ensuring a seamless and user-friendly experience across different devices and browsers.

3 Database Structure

Our database model and software architecture is not theoretical it **reflects the real-world practices used by central banks, clearinghouses** (payment platforms)**, and financial institutions worldwide**.

Modern interbank systems, such as **TARGET2 in Europe**, **Fedwire in the U.S.**, or **SWIFT's ISO 20022 migration**, rely on similar principles:

- A centralized message management system
- A standardized data format (ISO 20022)
- A high level of traceability, auditability, and security

Table 3.13: **detailed informations about the database**

Table Name	Purpose
BPDZDir	Contains registered banks and their essential identifiers (BIC, code banque, etc.)
BPDZHabi	Manages user accounts, access roles, and login credentials for each bank
BPDZCptBcDCA	Stores each bank's central account (Courant or Réserve) along with balances
BPDZmt	Central table for all ISO 20022 message movements (pacs.008, pacs.009, etc.)
BPDZPmtA	Archive of processed and finalized financial movements
BPDZMandat	Tracks payment mandates and their link to accounts and movements
BPDZPret	Handles bank loan requests and follow-up actions by authorized agents
BPDZSer	Registers available services by bank and supported message types
BPDZFraude	Stores blacklisted or flagged information for fraud prevention

This database is not just a repository, it is the **engine of the platform**, enabling seamless communication transaction logging, user control, and regulatory compliance. Designed with banking standards in mind, it reflects the needs of real financial institutions and provides a strong foundation for future development including automation, cross-border payments, and smart contract integration.

NOTE: Instead of creating a separate table for each ISO 20022 message type (pacs.008, pacs.009, camt.053, etc.), we centralized all messages into a single table named BPDZmt. This design decision brings several architectural and operational advantages: Better normalization, simplified SQL queries, improved extensibility, consistent status tracking.

NOTE: The word "movement" is used because it is a broad, industry-recognized term that refers to any kind of financial action between institutions. It supports multiple message types, aligns with ISO 20022 terminology, and enables future expansion without renaming or restructuring your system.

3.1 Example: Structure of the BPDZmt Table

```sql
CREATE TABLE BPDZmt (
    ID SERIAL PRIMARY KEY,
    StatutMouvement VARCHAR(2) CHECK (StatutMouvement IN ('99', '00','02','01','90','91','92','93','94','95','96','97','98')) NOT NULL DEFAULT '99',
    NatureMouvement VARCHAR(100) NOT NULL,
    ID_Banque_Emetteur INT ,
    ID_Banque_Recepteur INT ,
    TypeMessage VARCHAR(10) CHECK (TypeMessage IN ('Pacs.008', 'Pacs.009', 'Pacs.003', 'Pacs.004', 'camt.053')) NOT NULL,
    InformationsMessage TEXT,
    ModeTransmission VARCHAR(10) CHECK (ModeTransmission IN ('RTGS', 'CLRG', 'DNS')),
    XMLFile TEXT,
    DateCreation TIMESTAMP DEFAULT CURRENT_TIMESTAMP,
    DateReception TIMESTAMP DEFAULT CURRENT_TIMESTAMP,
    DateTraitement TIMESTAMP,
    FOREIGN KEY (ID_Banque_Emetteur) REFERENCES BPDZDir(ID) ON DELETE CASCADE,
    FOREIGN KEY (ID_Banque_Recepteur) REFERENCES BPDZDir(ID) ON DELETE CASCADE
);

CREATE INDEX idx_bpdzmt_emetteur ON BPDZmt(ID_Banque_Emetteur);
CREATE INDEX idx_bpdzmt_recepteur ON BPDZmt(ID_Banque_Recepteur);
CREATE INDEX idx_bpdzmt_typemessage ON BPDZmt(TypeMessage);
CREATE INDEX idx_bpdzmt_statutmouvement ON BPDZmt(StatutMouvement);
```

4 Implementation with Spring Boot

4.1 General Architecture

We adopted the **MVC (Model-View-Controller)** architecture to structure our Spring Boot application:

- **Model**: Represents entities such as BPDZmt, BPDZDir, and BPDZHabi.
- **Repository**: Uses JpaRepository for standard CRUD operations.
- **Service**: Contains the business logic for creating, processing, or archiving movements.
- **Controller**: Provides REST APIs to interact with the system externally.

4.2 Example: REST API in Spring Boot

```java
import com.bpdz.entity.BPDZmt;
import com.bpdz.service.PacsMessageService;
import org.springframework.http.ResponseEntity;
import org.springframework.web.bind.annotation.*;

@RestController
@RequestMapping("/api/pacs")
public class PacsController {

    private final PacsMessageService pacsMessageService;

    public PacsController(PacsMessageService pacsMessageService) {
        this.pacsMessageService = pacsMessageService;
    }

    @PostMapping("/008")
    public ResponseEntity<?> processPacs008(@RequestBody Pacs008DTO pacs008DTO) {
        try {
            BPDZmt movement = pacsMessageService.processPacs008(pacs008DTO);
            return ResponseEntity.ok(movement);
        } catch (Exception e) {
            return ResponseEntity.badRequest().body(e.getMessage());
        }
```

4.3 Transmission of PACS Messages via HTTP Using Configured Ports

To enable efficient interbank communication, our platform transmits ISO 20022 PACS messages (such as pacs.008, pacs.009) over HTTP. Each bank registered in the system has its own listening port, which is defined in the BPDZHabi table. This table manages not only user roles and authentication but also includes critical routing data such as IP addresses and port numbers assigned to each bank or participant.

This design allows the platform to dynamically determine the correct destination for each message. When a message is to be sent, the system queries the BPDZHabi table to retrieve the target bank's IP and port. It then opens a secure HTTP connection and transmits the XML-structured PACS message to the recipient bank's endpoint.

This method provides a lightweight and standardized transport mechanism that supports:

- Direct point-to-point communication between banks.
- Port-based routing, ensuring messages are sent to the correct institution.
- Interoperability, since all messages comply with ISO 20022 format.

This system architecture mimics real-world interbank communication standards while simplifying deployment and testing across institutions during development,we tested this using postman and Webhook.

4.4 Example: URLs in the Hardcode for Testing

```java
String senderCallbackUrl = "http://localhost:8081/api/pacs002";
String receiverCallbackUrl = "http://localhost:8082/api/pacs008";
```

5 SECURITY ASPECTS

5.1 User Access and Authentication

- **Encrypted Passwords**: All user passwords are stored in the database in a hashed form using a secure encryption algorithm (ByCrypt based on blowfish cypher encryption). This prevents the theft or retrieval of plain-text

passwords, even in the event of a data breach.

- **Role and Permission Management**: Through the BPDZHabi table, we defined different roles (Admin, Auditor, Viewer, etc.) with customized authorization levels. Access to specific operations is therefore restricted based on the user's role.
- **Automatic Session Timeout**: To strengthen protection against unauthorized access, an automatic session expiration feature has been implemented. If a user remains inactive for 10 minutes, the session is automatically terminated, protecting against misuse on unattended devices.

5.2 Protection Against Attacks

- **SQL Injection Prevention**: By using Spring Data JPA and parameterized queries, we effectively prevent SQL injection attacks. All interactions with the database are secured and validated.
- **Secure Spring Boot Framework**: One of the key reasons for choosing Spring Boot is its robustness and ability to provide a secure backend environment. With **Spring Security**, we integrated a strong protection layer on the backend, avoiding reliance on potentially weaker front-end protections.
- **Fraud Monitoring**: A dedicated table, BPDZFraude, is used to store blacklisted or suspicious information (such as bank names, BIC codes, countries, or account numbers). This mechanism helps identify high-risk transactions and can automatically block certain operations when needed.

6 Testing and Validation

Table 3.14: **detailed informations about the test**

Test Category	Status	Description
Unit Tests	Passed	Core logic, validation, formatting
Integration Tests	Passed	DB + REST communication verified
Functional API Tests	Passed	All endpoints handled valid/invalid input
Security Tests	Passed	SQL injection blocked, sessions handled
Consistency Tests	Passed	Foreign keys, data lifecycle validated

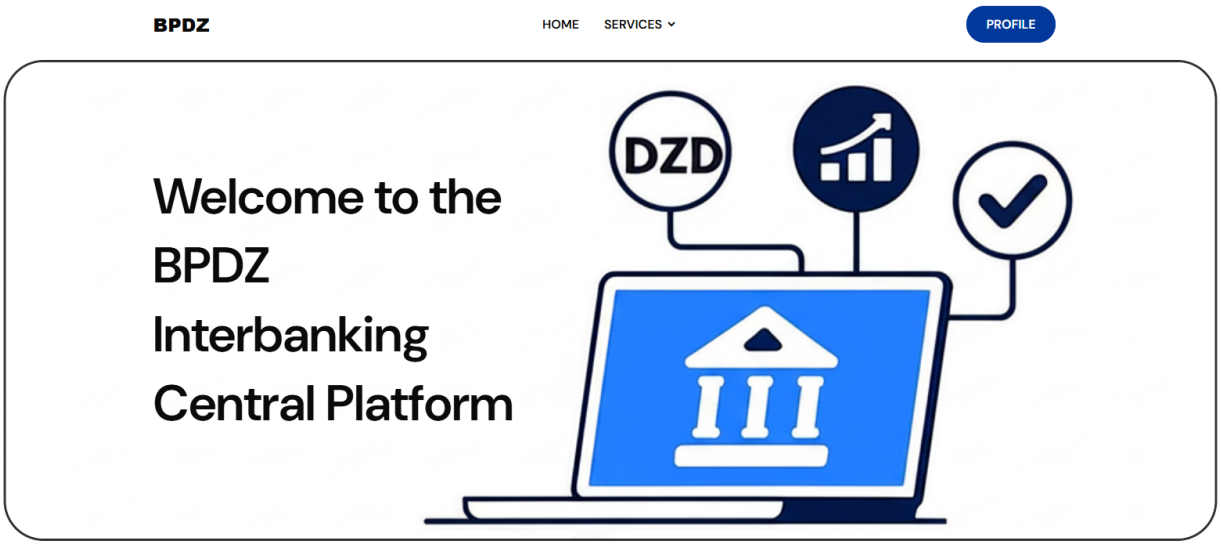


Figure 3.16: **Home page**

This interface displays BPDZ's minimalist welcome screen, featuring a clean greeting and intuitive navigation options. The uncluttered design prioritizes user accessibility and quick platform entry.

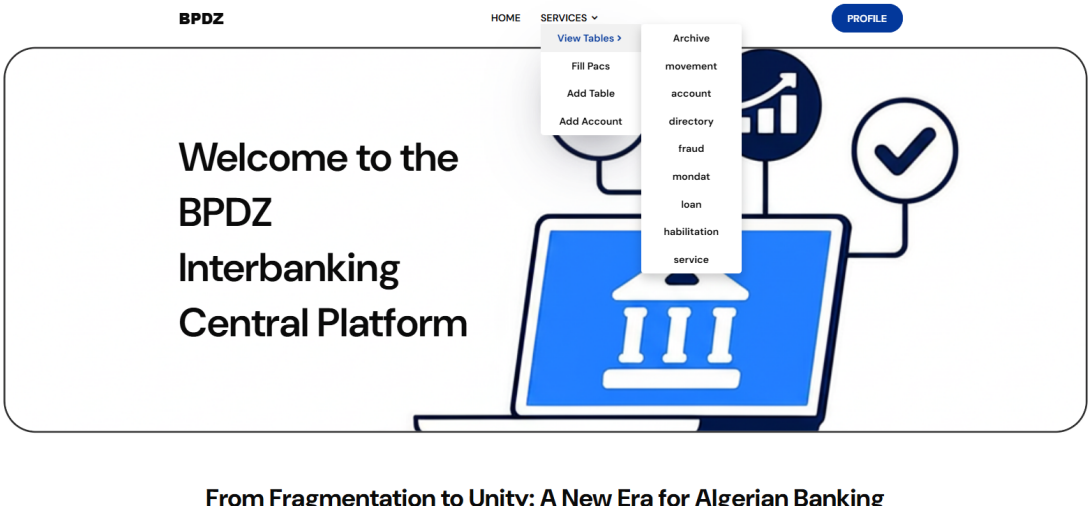


Figure 3.17: **Page dashboard**

This interface provides a streamlined BPDZ banking dashboard featuring essential tools for financial operations, including PACS form processing, account management, and fraud monitoring. Designed for efficiency, it combines real-time data tracking with ISO20022 compliance in a clean, user-friendly layout. The centralized platform enables secure administration of transactions and account oversight through intuitive controls.

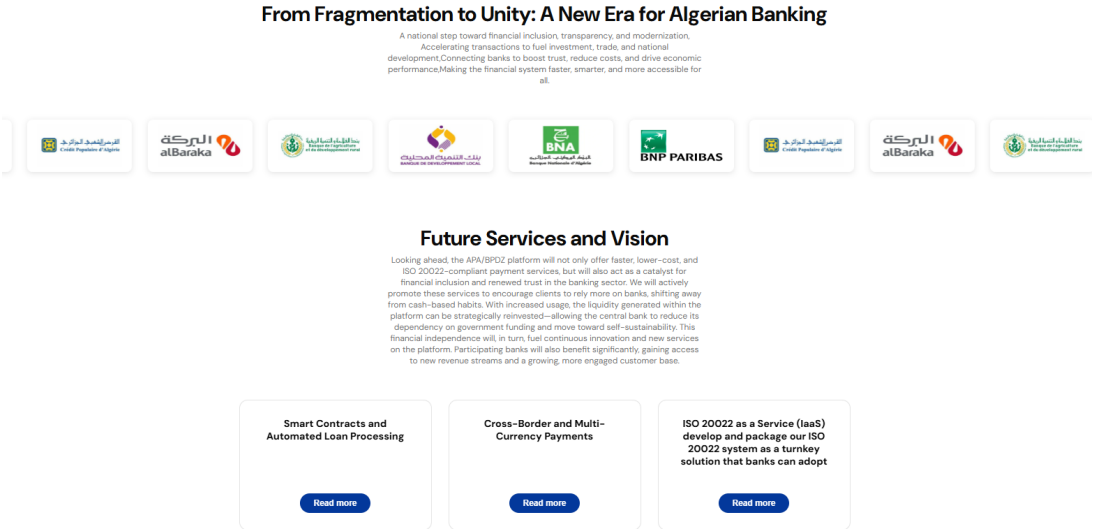


Figure 3.18: **Marketing for the platform**

This interface showcases BPDZ's banking solutions through strategic partnerships with institutions like BNA and BNP Paribas. The platform highlights secure, high-volume transaction processing with ISO 20022 compliance, featuring real-time analytics and bilingual Arabic/English support. Designed for financial professionals, it combines partnership visibility with functional demonstrations of our institutional-grade banking technology..

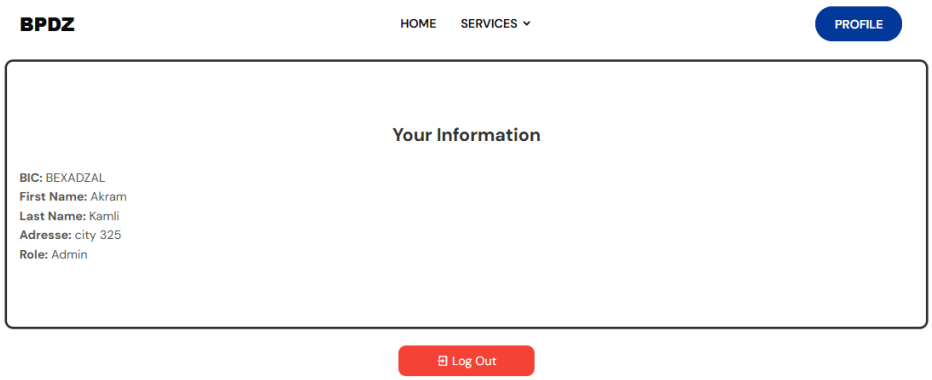


Figure 3.19: **Participant banks profiles**

This interface displays a BNA banker's profile within the BPDZ platform, showing essential details including BIC (BEXADZAL), user name (Akram Kamli), and branch address (AIN-BEDA).

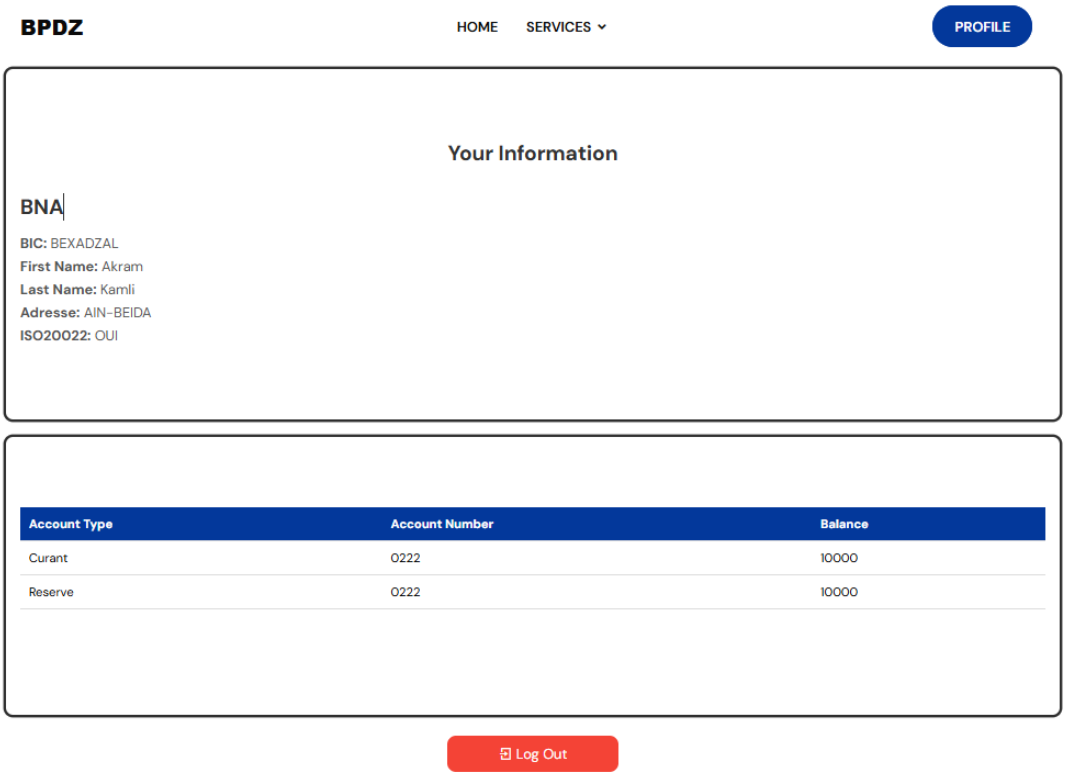


Figure 3.20: **Admins profile**

This interface displays a banker's profile within the BPDZ banking platform, showing user details (Akram Kamli, BIC: BEXADZAL, address: AIN-BEDA) with ISO20022 compliance enabled. The account summary lists two linked accounts (Current and Reserve, both 0222) with identical balance values. The clean layout prioritizes key banking information while providing quick access to logout.

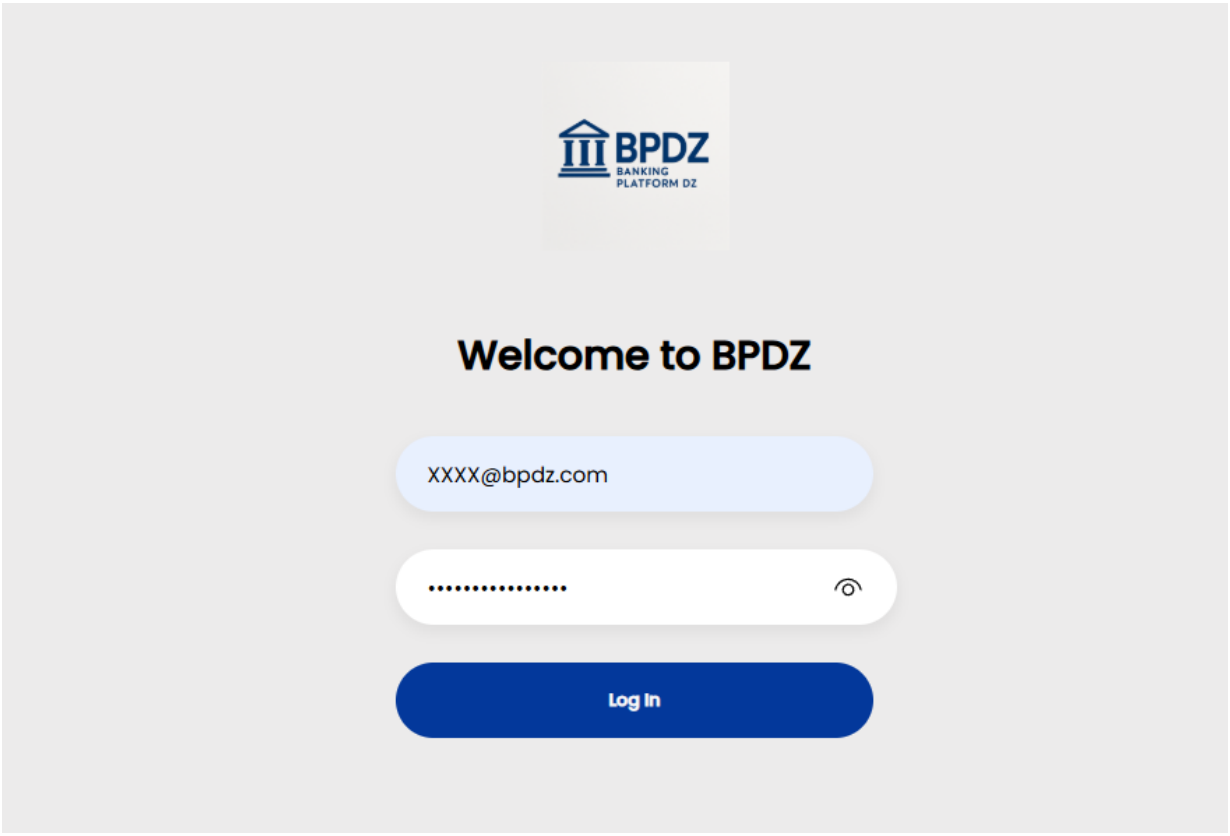


Figure 3.21: **Login**

This interface presents the BPDZ platform's clean login screen, featuring a simple welcome message and an email placeholder (XXXX@bpdz.com) above the login button. The minimalist design focuses on quick access while maintaining security.

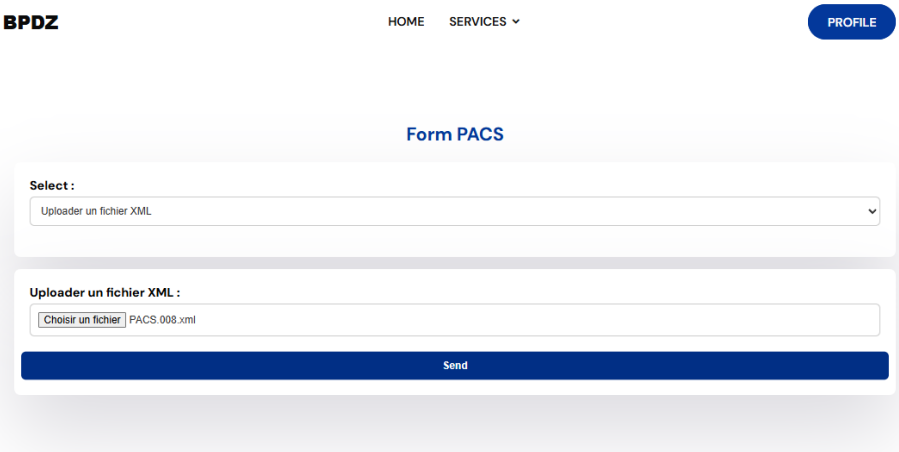


Figure 3.22: **Upload pacs**

This interface allows users to upload or manage PACS payment files in XML format. It provides options to choose an action, such as uploading a file, filling a form, or searching for an existing XML file. The screen includes a file selection button and a submission option to send the PACS 008 file.

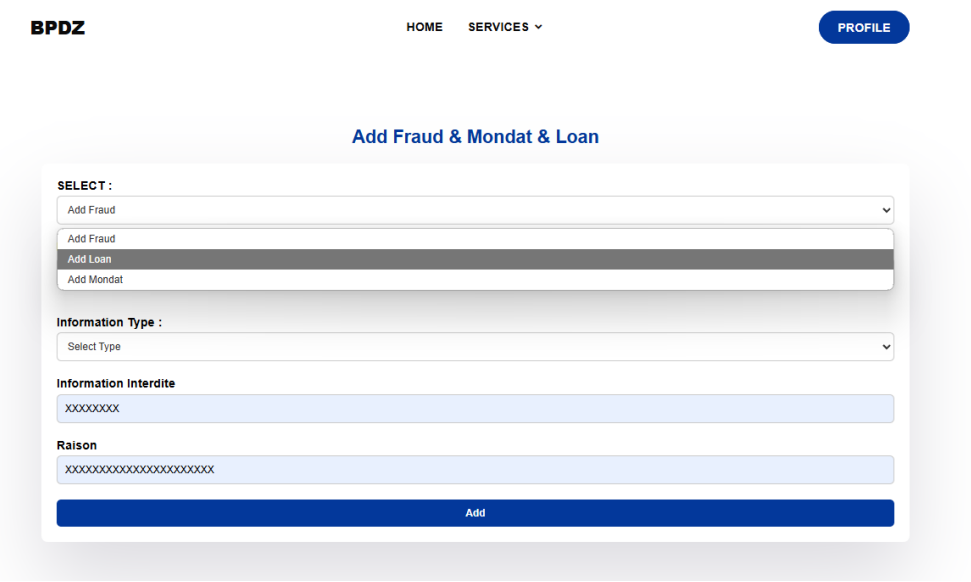


Figure 3.23: **Add loan mondat fraud**

This interface provides a financial operations screen for processing loans and payments within the BPDZ platform. It enables users to input loan details (amounts, issuance/coverage dates), manage payment networks, and record fraud cases, serving as a comprehensive tool for transaction management.

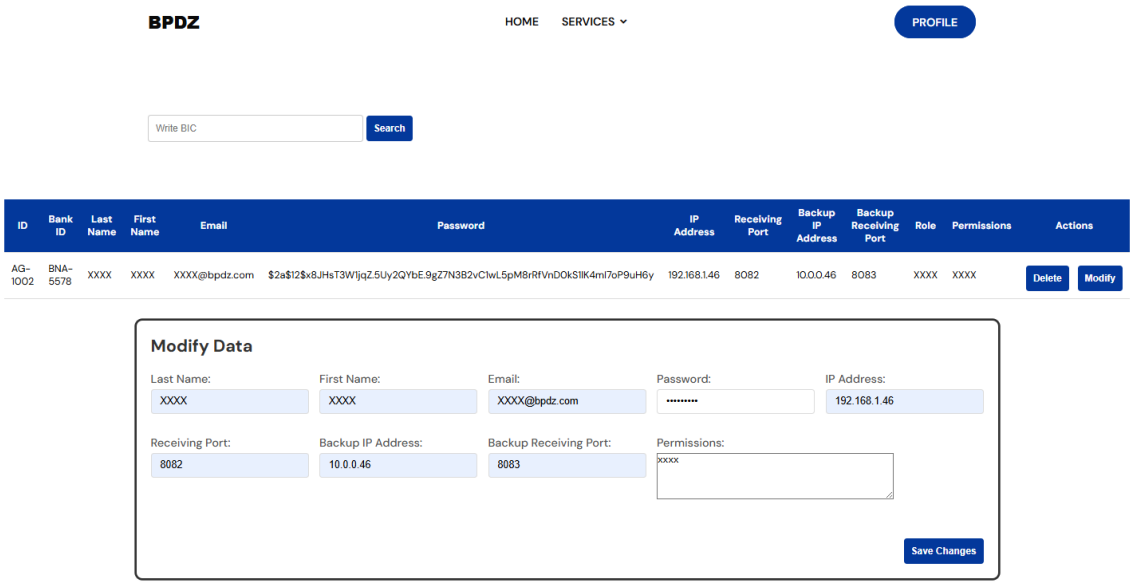


Figure 3.24: **Encryption**

This interface displays an administrative permissions management table, featuring user credentials with encrypted passwords, associated IP addresses, access level details, and administrative action options like ”Delete” or ”Change” for privilege control. The structured layout enables secure user privilege management through clear visibility of access rights and modification capabilities.

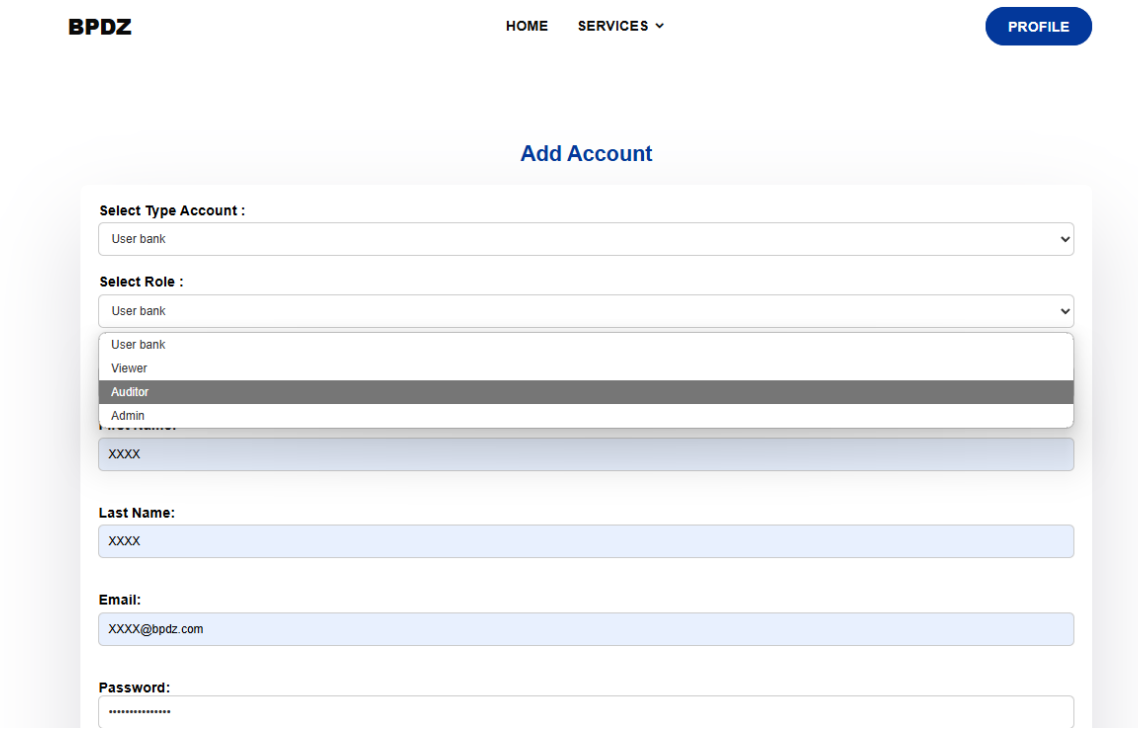


Figure 3.25: **Add account**

This interface allows users to add a new bank account in the BPDZ system. It collects essential details such as BIC, first name, last name, and email, along with bank selection options. The form ensures proper account setup with structured input fields.

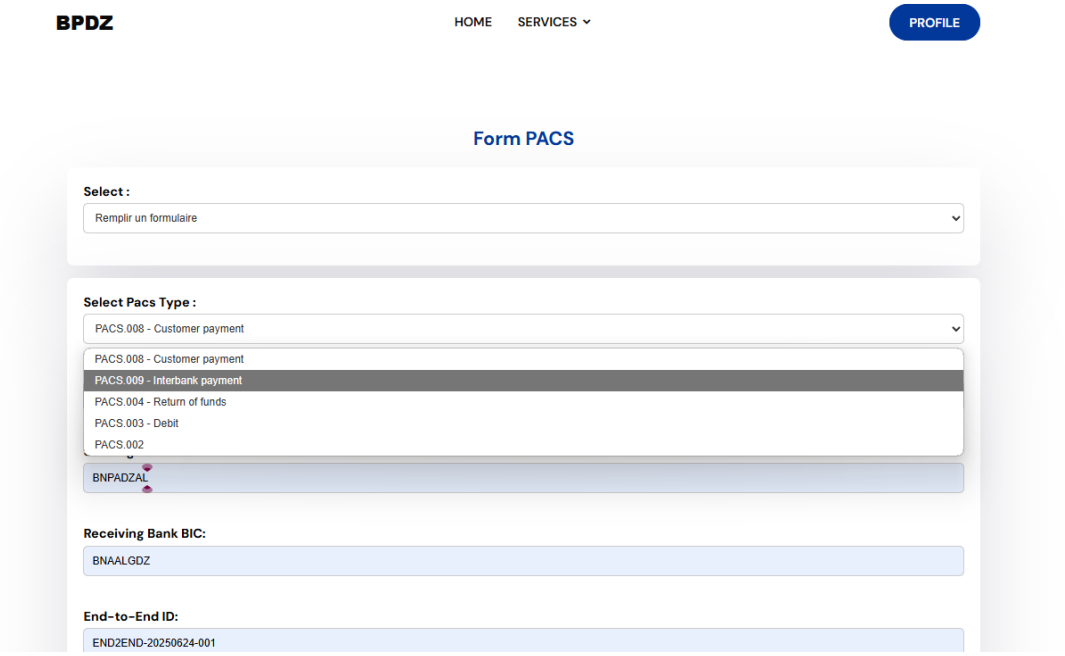


Figure 3.26: **fill forma**

This interface presents the BPDZ banking platform's PACS transaction form, enabling users to select payment types (e.g., PACS.008 for customer payments) and enter essential details like beneficiary BIC ("CDEFA/PNN") and Instruction ID ("2135543"). The partially completed form, with empty fields such as "End to End ID," features a structured layout with dropdown menus and clear labeling to streamline payment processing.

7 Conclusion

This banking application represents a comprehensive implementation of modern interbank communication standards, successfully bridging the gap between theoretical financial protocols and practical, real world banking operations. By leveraging Spring Boot's robust framework and PostgreSQL's reliable database capabilities, we've created a system that authentically mirrors the operational models of central banks and financial institutions worldwide.

The project's core achievement lies in its innovative unified table model for ISO 20022 message processing, which simplifies integration while maintaining the rigorous standards required for financial transactions. Our design choicesfrom the centralized BPDZmt table to port based HTTP communication demonstrate a thoughtful balance between banking industry conventions and technical efficiency.

Security has been implemented as a fundamental architecture principle rather than an afterthought, with multiple protective layers including encrypted authentication, role based access control, and comprehensive fraud monitoring. The successful completion of all test categories confirms the system's reliability and readiness for production environments.

Looking ahead, the platform is well positioned for expansion, with clear pathways for incorporating additional ISO 20022 message types, cross border payment capabilities, and even smart contract functionality. The vision of offering ISO 20022 as a Service (IaaS) promises to make these banking technologies more accessible to financial institutions of all sizes.

This project not only delivers immediate value as a working banking platform but also establishes a foundation for the future of financial technology in Algeria and potentially across the African and Arab regions. By combining international standards with local requirements, we've created a system that can grow alongside the evolving needs of modern banking while maintaining strict compliance, security, and operational efficiency.

1 General conclusion

The implementation of the **APA (Algerian Payment Area)** platform, supported by the **BPDZ** system and structured around **ISO 20022 standards**, offers a concrete and technically feasible solution to Algeria's fragmented interbank ecosystem. Through this project, we demonstrated that a **centralized**, **secure**, and **standardized** platform can effectively address the major weaknesses identified in Algeria's financial infrastructure:

- **Real-time payments and traceability** are achieved via ISO 20022 messages such as pacs.008, pacs.009, and camt.053, ensuring secure, fast, and fully transparent transactions.
- **System interoperability** is enabled by a unified message model and a centralized database (BPDZmt), allowing all banks regardless of their internal systems to communicate using the same format and logic.
- **Security and compliance** are enforced by using robust technologies like Spring Boot, PostgreSQL, encrypted user authentication, and fraud detection mechanisms (e.g., BPDZFraude).
- **Operational scalability** is guaranteed by our modular architecture, which supports future integration of smart contracts, cross-border payments, and additional ISO 20022 messages (e.g., pain.001, pacs.010).
- **Economic impact** is addressed through a detailed ROI study, demonstrating that the platform becomes profitable in the medium term and highly beneficial in the long term, especially when adopted by public institutions and integrated into national monetary strategy.

Ultimately, this report proves that the proposed APA platform is **not only a technical prototype** but also a **strategic framework** that, if implemented by the Bank of Algeria and supported by local banks, could radically improve financial inclusion, reduce cash dependency, and align Algeria's banking sector with international standards and future regional integration goals.

2 Project Limitations

While the proposed APA platform offers numerous benefits, certain limitations must be acknowledged:

- **Institutional Challenges**: The successful implementation of the APA requires strong coordination among various stakeholders, including banks, regulatory bodies, and government institutions.
- **Technical Constraints**: Integrating ISO 20022 standards and ensuring interoperability with existing systems may present technical challenges.
- **Financial Considerations**: The initial investment required for infrastructure development and system integration is substantial.
- **Regulatory Framework**: Establishing a comprehensive legal and regulatory framework is essential to govern the operations of the APA platform.

3 FUTURE WORK AND PERSPECTIVES

3.1 Integration of Additional ISO 20022 Messages

One of the next major steps is the **integration of additional ISO 20022 messages**, including:

- pacs.010: For financial institution credit transfers.
- camt.054: For debit/credit notification messages.
- pain.001 and pain.002: For customer-to-bank payment initiation and status reports.

Adding support for these messages will allow the platform to **fully support the payment lifecycle**, from initiation to settlement and reconciliation.

3.2 Smart Contracts and Automated Loan Processing

We plan to incorporate **smart contracts**, especially in the context of loans and credit agreements. This involves:

- Automated loan disbursement and repayment.
- Interest calculation and penalty management.
- SWIFT-like secure contracts.

3.3 Cross-Border and Multi-Currency Payments

To support Algeria's ambition to become a financial bridge across Africa and the Arab region, we plan to develop:

- Multi-currency payment handling.
- Cross-border compliance modules.
- SWIFT GPI-style tracking.

3.4 Web-Based Monitoring Interface and Advanced Analytics

We also envision the development of:

- A responsive web dashboard.
- Advanced analytics.
- Exportable audit logs and traceability tools compliant with financial regulations.
- **Advanced analytics** modules to support macroeconomic monitoring by the central bank.

3.5 ISO 20022 as a Service (IaaS)

In the future, our goal is to **develop and package our ISO 20022 system as a turnkey solution** that banks can adopt with minimal internal development effort. Instead of building their own infrastructure from scratch.

4 Recommendations for Real-World Implementation

To address these limitations and ensure the successful implementation of the APA platform, the following recommendations are proposed:

1. **Stakeholder Engagement**: Foster collaboration among banks, fintech companies, regulatory authorities, and government agencies to align objectives and resources.
2. **Phased Implementation**: Adopt a gradual rollout strategy, starting with pilot programs to test functionalities and gather feedback.
3. **Capacity Building**: Invest in training and development programs to equip stakeholders with the necessary skills and knowledge.
4. **Regulatory Development**: Establish clear legal and regulatory frameworks to support the operations and governance of the APA platform.
5. **Public Awareness**: Conduct awareness campaigns to educate the public and promote the adoption of digital payment systems.

By addressing these recommendations, Algeria can pave the way for a resilient, inclusive, and modern interbank payment system that supports its economic growth and integration into the global financial landscape.

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